Chapter 13

Project Management: PERT and CPM

"Successful project management requires more front and back end resources (and less middle) than are usually allocated."

– Anonymous

PREVIEW

The objective of project management study is to schedule activities associated with any project in an efficient manner so as to complete the project on or before a specified time limit and at the minimum cost with specified quality standard.

Two project management techniques – PERT and CPM are commonly used to show the logical sequence of activities to be performed in any project in order to achieve project objectives.

LEARNING OBJECTIVES

After studying this chapter you should be able to

- understand the significance of using PERT and CPM techniques for project management.
- know the basic difference between PERT and CPM network techniques.
- understand phases of any project and various activities that need to be done during these phases.
- construct network diagrams with single and three time estimates of activities involved in a project.
- determine critical path and floats associated with non-critical activities and events along with total project completion time.
- determine the probability of completing a project on or before the schedule date.
- know how to update a project along with resource leveling and smoothing.
- crash project schedule time and establish a time-cost trade-off for the completion of a project.

CHAPTER OUTLINE

13.1 Introduction
13.2 Basic Difference Between Pert and Cpm
13.3 Phases of Project Management
13.4 Pert/Cpm Network Components and Precedence Relationships
- Conceptual Questions A
- Self Practice Problems A

13.5 Critical Path Analysis
- Conceptual Questions B
- Self Practice Problems B
- Hints and Answers

13.6 Project Scheduling with Uncertain Activity Times
- Conceptual Questions C
- Self Practice Problems C
- Hints and Answers

13.7 Project Time-Cost Trade-Off
- Self Practice Problems D
- Hints and Answers

13.8 Updating of the Project Progress
13.9 Resource Allocation
- Self Practice Problems E

❑ Chapter Summary
❑ Chapter Concepts Quiz
❑ Case Study

13.1 INTRODUCTION

A project involves a large number of interrelated *activities* (*or tasks*) that must be completed on or before a specified time limit, in a specified sequence (or order) with specified quality and minimum cost of using *resources* such as personnel, money, materials, facilities and/or space. Examples of projects include, construction of a bridge, highway, power plant, repair and maintenance of an oil refinery or an air plane; design, development and marketing of a new product, research and development work, etc. Since a project involves large number of interrelated activities, therefore it is necessary to prepare a plan for scheduling and controlling these activities (or tasks). This approach will help in identifying bottlenecks and even discovering alternate work-plan for the project.

Network Analysis, Network Planning or Network Planning and Scheduling Techniques are used for planning, scheduling and controlling large and complex projects. These techniques are based on the representation of the project as a network of activities. A network is a graphical presentation of arrows and nodes for showing the logical sequence of various activities to be performed to achieve project objectives. In this chapter, we shall discuss two of these well-known techniques – PERT and CPM.

PERT/CPM is a technique used for assisting project managers carry out their responsibilities.

PERT (*Programme Evaluation and Review Technique*) was developed in 1956–58 by a research team to help in the planning and scheduling of the US Navy's Polaris Nuclear Submarine Missile project involving thousands of activities. The objective of the team was to efficiently plan and develop the Polaris missile system. This technique has proved to be useful for projects that have an element of uncertainty in the estimation of activity duration, as is the case with new types of projects which have never been taken up before.

CPM (*Critical Path Method*) was developed by E.I. DuPont company along with Remington Rand Corporation almost at the same time, 1956-58. The objective of the company was to develop a technique to monitor the maintenance of its chemical plants. This technique has proved to be useful for developing time-cost trade-off for projects that involve activities of repetitive nature.

13.2 BASIC DIFFERENCE BETWEEN PERT AND CPM

Both PERT and CPM share in common the determination of a critical path and are based on the network representation of activities and their scheduling that determines the most critical activities to be controlled so as to meet the completion date of a project. However, the following are some of their major differences.

PERT

1. In PERT analysis, a weighted average of the expected completion time of each activity is calculated given three time estimates of its completion. These time estimates are derived from probability distribution of completion times of an activity.
2. In PERT analysis emphasis is given on the completion of a task rather than the activities required to be performed to complete a task. Thus, PERT is also called *an event-oriented technique*.
3. PERT is used for one time projects that involve activities of non-repetitive nature (i.e. activities that may never have been performed before), where completion times are uncertain.
4. PERT helps in identifying critical areas in a project so that necessary adjustments can be made to meet the scheduled completion date of the project.

CPM

1. In CPM, the completion time of each activity is known with certainty that too unique.
2. CPM analysis explicitly estimate the cost of the project in addition to the completion time. Thus, this technique is suitable for establishing a trade-off for optimum balancing between schedule time and cost of the project.
3. CPM is used for completing of projects that involve activities of repetitive nature.

13.2.1 Significance of Using PERT/CPM

1. A network diagram helps to translate complex project into a set of simple and logical arranged activities and therefore,
   - helps in the clarity of thoughts and actions.

- helps in clear and unambiguous communication developing from top to bottom and vice versa among the people responsible for executing the project.

2. Detailed analysis of a network helps project incharge to peep into the future because
   - difficulties and problems that can be reasonably expected to crop up during the course of execution, can be foreseen well ahead of its actual execution.
   - delays and holdups during course of execution are minimized. Corrective action can also be taken well in time.
3. Isolates activities that control the project completion and therefore, results in expeditious completion of the project.
4. Helps in the division of responsibilities and therefore, enhance effective coordination among different departments/agencies involved.
5. Helps in timely allocation of resources to various activities in order to achieve optimal utilization of resources.

13.3 PHASES OF PROJECT MANAGEMENT

In general, project management consists of three phases: Planning, Scheduling and Control.

1. Project planning phase In order to understand the sequencing or precedence relationship among activities in a project, it is essential to draw a network diagram. The steps involved during this phase are listed below:

(i) Identify various activities (tasks or work packages/elements) to be performed in the project, that is, develop a breakdown structure (WBS).

(ii) Determine the requirement of resources such as men, materials, machines, money, etc., for carrying out activities listed above.

(iii) Assign responsibility for each work package. The work packages corresponds to the smallest work efforts defined in a project and forms the set of tasks that are the basis for planning, scheduling and controlling the project.

(iv) Allocate resources to work packages.

(v) Estimate cost and time at various levels of project completion.

(vi) Develop work performance criteria.

(vii) Establish control channels for project personnel.

2. Scheduling phase Once all activities have been identified and given unique codes, the project scheduling (when each of the activities is required to be performed) is taken up. Prepare an estimate of the likelihood of the project to be completed on or before the specified time. The steps involved during this phase are listed below:

(i) Identify all people who will be responsible for each task.

(ii) Estimate the expected duration(s) of each activity, taking into consideration the resources required for their execution in the most economic manner.

(iii) Specify the interrelationship (i.e. precedence relationship) among various activities.

(iv) Develop a network diagram, showing the sequential interrelationship between various activities. For this, tips such as; *what* is required to be done; *why* it must be done, can it be dispensed with; *how* to carry out the job; *what* must precede it; *what* has to follow; *what* can be done concurrently, may be followed.

(v) Based on these time estimates, calculate the total project duration, identify critical path; calculate floats; carry out resources smoothing (or levelling) exercise for critical (or scare) resources, taking into account the resource constraints (if any).

3. Project control phase Project control refers to the evaluation of the actual progress (status) against the plan. If significant differences are observed, then remedial (modifying planning) or reallocation of resources measures are adopted in order to update and revise the uncompleted part of the project.

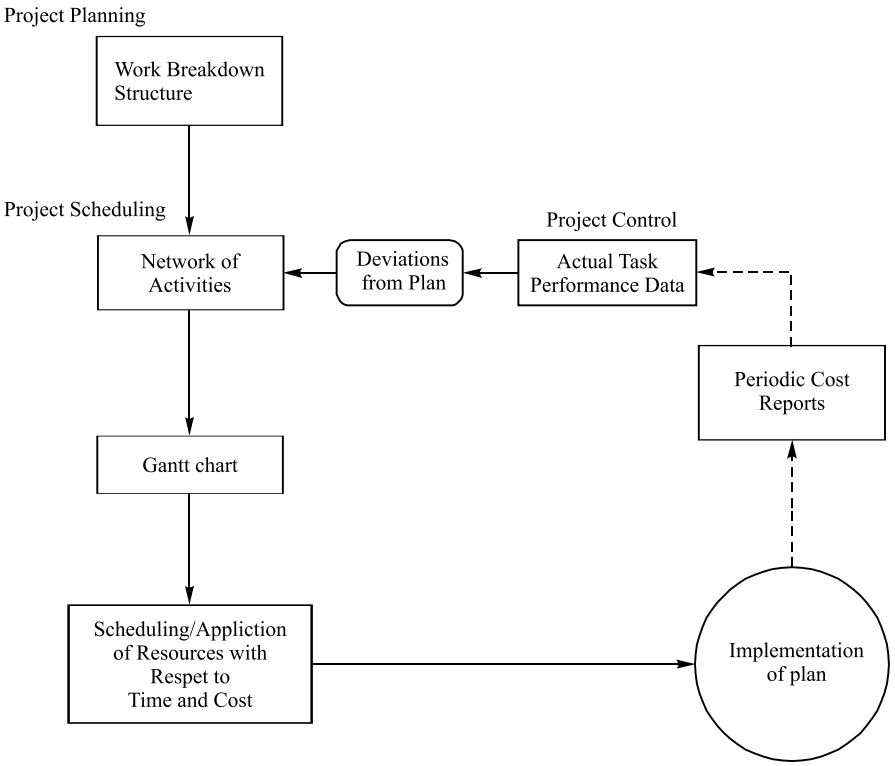


Fig. 13.1 Relationship between Phases of Project Management

The relationship among phases of project management is shown in Fig. 13.1. It indicates how the actual task performance data is used to track deviations from the original plan and schedule. These adjustments are helpful to aggregate work packages into subsystems and track the progress of these subsystems as part of the reporting and review procedures.

13.4 PERT/CPM NETWORK COMPONENTS AND PRECEDENCE RELATIONSHIPS

PERT/CPM network consists of two major components. These are discussed below:

Events Events in the network diagram represent project milestones, such as the start or the completion of an activity (task) or activities, and occur at a particular instant of time at which some specific part of the project has been or is to be achieved. Events are commonly represented by circles (nodes) in the network diagram.

The events can be further classified into the following two categories:

(i) *Merge Event* : An event which represents the joint completion of more than one activity is known as a merge event. This is shown in Fig. 13.2(a).



(a) Merge event (b) Burst event

Fig. 13.2 Types of Events

(ii) *Burst Event* : An event that represents the initiation (beginning) of more than one activity is known as burst event. This is shown in Fig. 13.2(b).

Events in the network diagram are identified by numbers. Each event should be identified by a number higher than that the one allotted to its immediately preceding event to indicate progress of work. The

numbering of events in the network diagram must start from left (start of the project) to the right (completion of the project) and top to the bottom. Care should be taken that there is no duplication in the numbering.

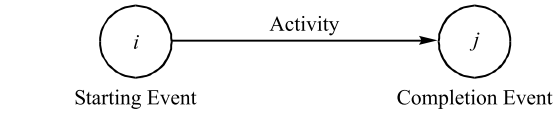
Activities Activities in the network diagram represent project operations (or tasks) to be conducted. As such each activity except dummy activity requires resources and takes a certain amount of time for completion. An arrow is commonly used to represent an activity with its head indicating the direction of progress in the project.

Activities are identified by the numbers of their starting (tail or initial) event and ending (head, or terminal) event, for example, an arrow (i, j) between two events; the tail event i represents the start of the activity and the head event j represents the completion of the activity as shown in Fig. 13.3(a). The activities can be further classified into the following three categories:

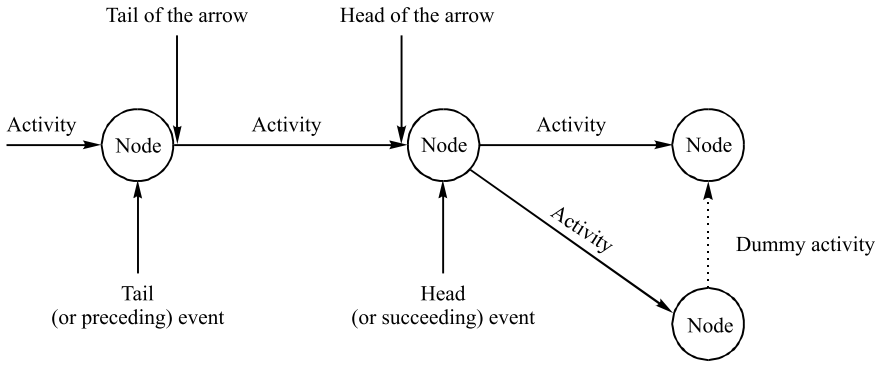
Activity is a distinct task that needs to be performed as part of a project.

(i) *Predecessor Activity:* An activity which must be completed before one or more other activities start is known as predecessor activity.
(ii) *Successor Activity:* An activity which starts immediately after one or more of other activities are completed is known as successor activity.
(iii) *Dummy Activity:* An activity which does not consume either any resource and/or time is known as dummy activity.

A dummy activity in the network is added only to establish the given precedence relationship among activities of the project. It is needed when (a) two or more parallel activities in a project have same head and tail events, or (b) two or more activities have some (but not all) of their immediate predecessor activities in common. A dummy activity is shown by a dotted line in the network diagram as shown in Fig. 13.3(b).



(a) Activity-Node Relationship



(b) Activity-Node Relationship

AON project network where each activity is represented by a node (circle or rectangle); the arrows (arcs) show the precedence relationships between the activities.

Fig. 13.3 Activity-Node Relationship in Network Diagram

Network models use the following two types of precedence network to show precedence requirements of the activities in the project.

Activity-on-Node (AON) network In this type of precedence network each node (or circle) represents a specific task while the arcs represent the ordering between tasks. AON network diagrams place the activities within the nodes, and the arrows are used to indicate sequencing requirements. Generally, these diagrams have no particular starting and ending nodes for the whole project. The lack of dummy activities in these diagrams always make them easier to draw and to interpret.

Activity-on-Arrow (AOA) network In this type of precedence network at each end of the activity arrow is a node (or circle). These nodes represent points in time or instants, when an activity is starting or ending. The arrow itself represents the passage of time required for that activity to be performed.

These diagrams have a single beginning node from which all activities with no predecessors may start. The diagram then works its way from left to right, ending with a single ending node, where all activities with no followers come together. Three important advantages of using AOA are as follows:

(i) Many computer programs are based on AOA network.
(ii) AOA diagrams can be superimposed on a time scale with the arrows drawn, the correct length to indicate the time requirement.
(iii) AOA diagrams give a better sense of the flow of time throughout a project.

In **AOA project network** each activity is represented by an arrow (or arc).

In this chapter, only AOA network diagrams will be used. Fig. 13.4 illustrates AON and AOA diagramming.

13.4.1 Rules for AOA Network Construction

Following are some of the rules that have to be followed while constructing a network:

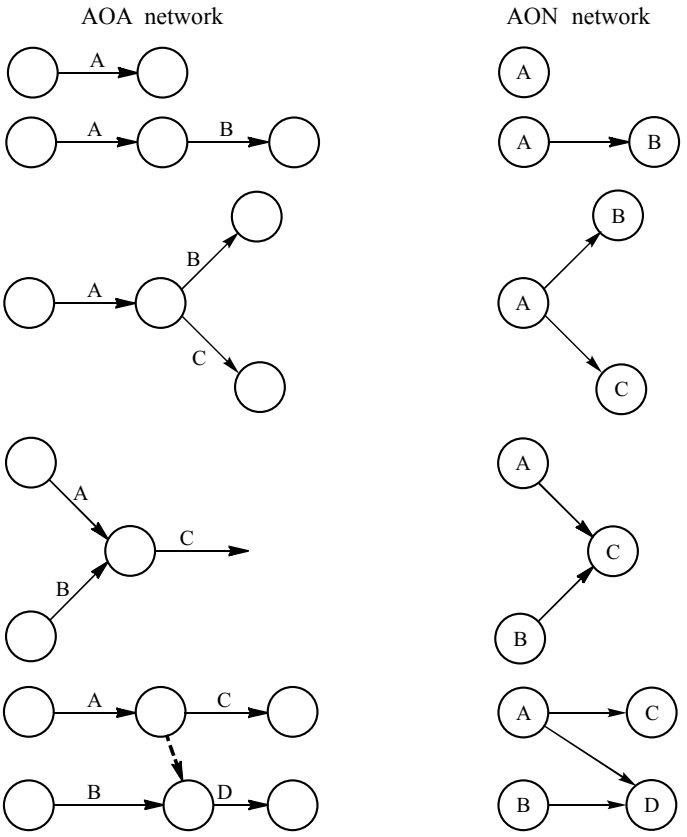


Fig. 13.4 AOA and AON Diagramming

1. In network diagram, arrows represent activities and circles the events. The length of an arrow is of no significance.
2. Each activity should be represented only by one arrow and must start and end in a circle called *event*. The tail of an activity represents the start, and head the completion of work.
3. The event numbered 1 denotes the start of the project and is called *initial event*. All activities emerging (or taking off) from event 1 should not be preceded by any other activity or activities. An event carrying the highest number denotes the completion event. A network should have only one initial event and only one terminal event.
4. The general rule for numbering the event is that the head event should always be numbered larger than the number at its tail. That is, events should be numbered such that for each activity (i, j), $i < j$.
5. An activity must be uniquely identified by its starting and completion event, which implies that:
   (a) An event number should not get repeated or duplicated.
   (b) Two activities should not be identified by the same completion event.
   (c) Activities must be represented either by their symbols or by the corresponding ordered pair of starting-completion events.
6. The logical sequence (or interrelationship) between activities must follow following rules:
   (a) An event cannot occur until all its incoming activities have been completed.
   (b) An activity cannot start unless all the preceding activities, on which it depends, have been completed.

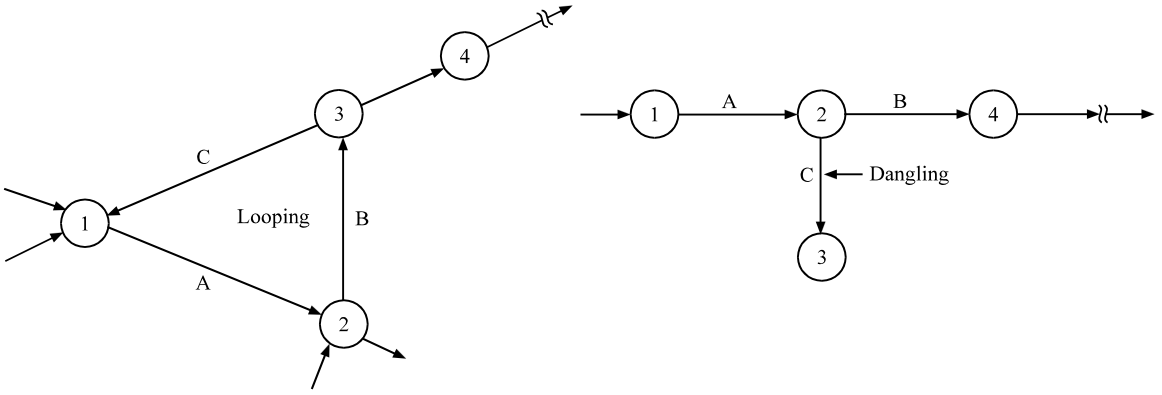
(c) Though a dummy activity does not consume either any resource of time, even then it has to follow the rules 6(a) and (b).

13.4.2 Errors and Dummies in Network

Looping and Dangling Looping (cycling) and dangling are considered as faults in a network. Therefore, these must be avoided.

(i) A case of endless loop in a network diagram, which is also known as *looping,* is shown in Fig. 13.5(a), where activities *A*, *B* and *C* form a cycle.

Due to precedence relationships, it appears from Fig. 13.5(a) that every activity in looping (or cycle) is a predecessor of itself. In this case it is difficult to number three events associated with activity *A, B* and *C* so as to satisfy rule 6 of constructing the network.



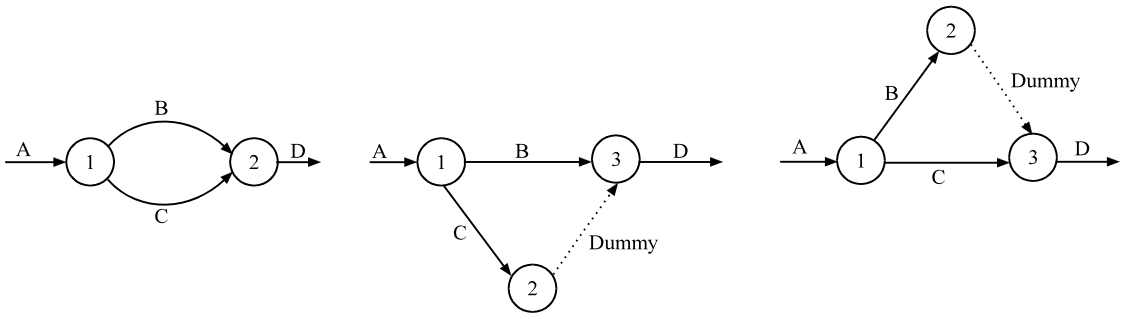
(a) Looping (b) Dangling

Fig. 13.5 Looping and Dangling in the Network Diagram

(ii) A case of disconnect activity before the completion of all activities, which is also known as *dangling*, is shown in Fig. 13.5(b). In this case, activity *C* does not give any result as per the rules of the network. The dangling may be avoided by adopting rule 5 of constructing the network.

Dummy* (*or Redundant*) *Activity The following are the two cases in which the use of dummy activity may help in drawing the network correctly, as per the various rules.

(i) When two or more parallel activities in a project have the same head and tail events, i.e. two events are connected with more than one arrow.



(a) Parallel Activities (b) Dummy Activity (c) Dummy Activity

Fig. 13.6 Parallel Activities and Dummy Activity

In Fig. 13.6(a), activities *B* and *C* have a common predecessor – activity *A.* At the same time, they have activity *D* as a common successor. To arrive correct network, a dummy activity for the ending event *B* to show that *D* may not start before *B* and *C*, is completed. This is shown in Fig. 13.6(b).

(ii) When two chains of activities have a common event, yet are completely or partly independent of each other, as shown in Fig. 13.7(a). A dummy which is used in such a case, to establish proper logical relationships, is also known as *logic dummy activity.*

In Fig. 13.7(a), if head event of *C* and *D* do not depend on the completion of activities *A* and *B*, then the network can be redrawn, as shown in Fig. 13.7(b). Otherwise, the pattern of Fig. 13.7(a) must be followed:

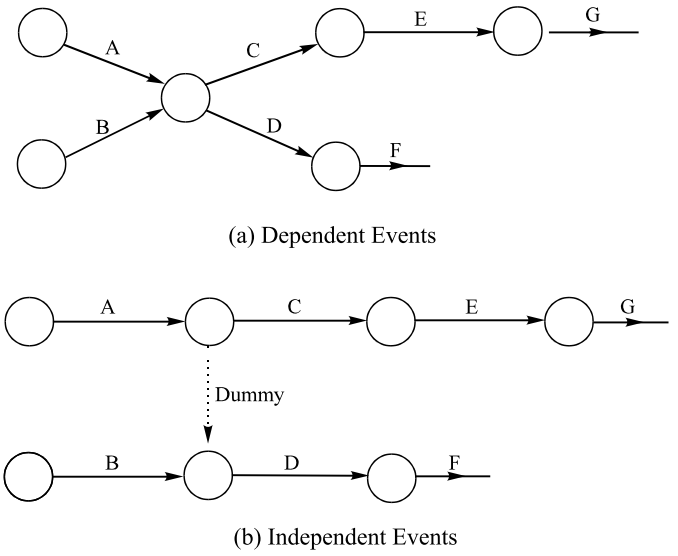


Fig. 13.7 Dependent and Independent Events

Example 13.1 An assembly is to be made from two parts *X* and *Y*. Both parts must be turned on a lathe. *Y* must be polished whereas *X* need not be polished. The sequence of activities, together with their predecessors, is given below.

Activity	*Description*	*Predecessor Activity*
A	Open work order	–
B	Get material for *X*	*A*
C	Get material for *Y*	*A*
D	Turn *X* on lathe	*B*
E	Turn *Y* on lathe	*B, C*
F	Polish *Y*	*E*
G	Assemble *X* and *Y*	*D, F*
H	Pack	*G*

Draw a network diagram of activities for the project.

Solution The network diagram for the project is shown in Fig. 13.8.

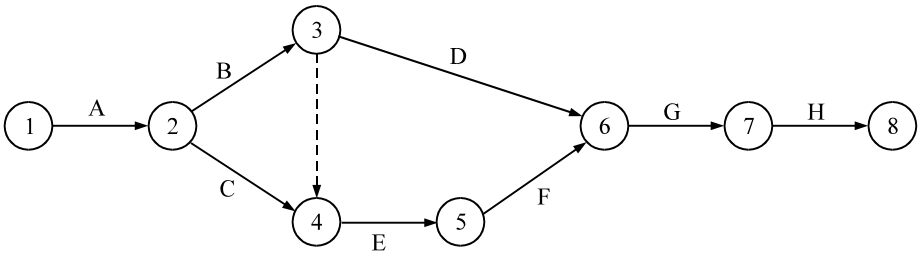


Fig. 13.8 Network Diagram

Example 13.2 Listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery.

Activity	*Description*	*Predecessor Activity*
A	Dismantle pipe connections	—
B	Dismantle heater, closure, and floating front	*A*
C	Remove tube bundle	*B*
D	Clean bolts	*B*
E	Clean heater and floating head front	*B*
F	Clean tube bundle	*C*
G	Clean shell	*C*
H	Replace tube bundle	*F, G*
I	Prepare shell pressure test	*D, E, H*
J	Prepare tube pressure test and reassemble	*I*

Draw a network diagram of activities for the project.

Solution The network diagram for the project is shown in Fig. 13.9.

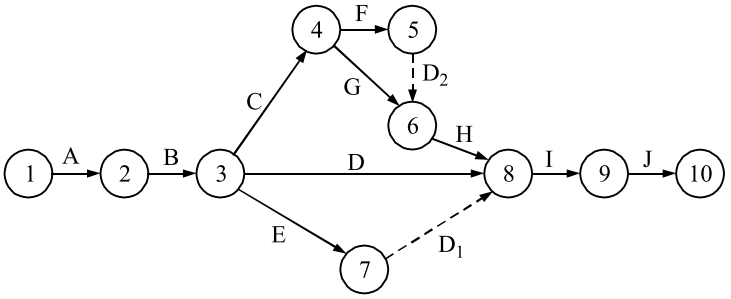


Fig. 13.9
Network Diagram

Example 13.3 Listed in the table are the activities and sequencing necessary for the completion of a recruitment procedure for management trainees (MT) in an organization.

Activity	*Description*	*Predecessor Activity*
A	Asking units for requirements	–
B	Ascertaining management trainees (MTs) requirements for commercial functions	*A*
C	Ascertaining MTs requirement for Accounts/Finance functions	*A*
D	Formulating advertisement for MT(A/C)	*C*
E	Formulating advertisement for MT (Commercial)	*B*
F	Calling applications from the successful candidates passing through the Institute of Chartered Accountants (ICA)	*C*
G	Releasing the advertisement	*D, E*
H	Completing applications received	*G*
I	Screening of applications against advertisement	*H*
J	Screening of applications received from ICA	*F*
K	Sending of personal forms	*I, J*
L	Issuing interview/regret letters	*K*
M	Preliminary interviews	*L*
N	Preliminary interviews of outstanding candidates from ICA	*J*
O	Final interview	*M, N*

Draw a network diagram of activities for the project.

Solution The network diagram for the project is shown in Fig. 13.10.

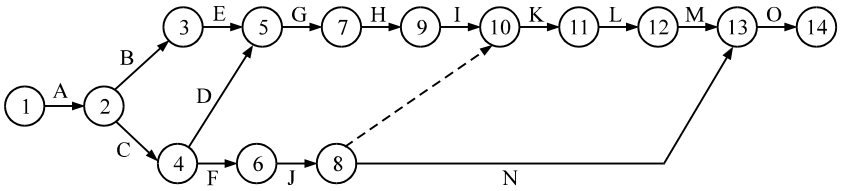


Fig. 13.10
Network Diagram

CONCEPTUAL QUESTIONS A

1. Briefly mention the areas of application of network techniques.
2. Compare and contrast CPM and PERT. Under what conditions would you recommend the scheduling by PERT? Justify your answer with reasons.
3. Discuss the various steps involved in the applications of PERT and CPM.
4. Highlight the difficulties encountered in using network techniques.
5. Explain the basic logic of arrow networks.
6. Explain the reasons for incorporating dummy activities in a network diagram. In what way do these differ from the normal activities?
7. State the circumstances where CPM is a better technique of project management than PERT. [*Delhi Univ., MCom, 2002*]
8. How does the PERT technique help a business manager in decision-making? [*Delhi Univ., MCom, 2000*]
9. Critically comment on the assumptions that are mode for PERT/CPM analysis of projects.
10. PERT now has proven to be an effective management tool that can be utilized by companies of all sizes in almost every industry. Discuss. [*Delhi Univ., MBA, 2001*]
11. 'PERT provides the framework with which a project can be described, scheduled and then controlled?' Discuss. [*Delhi Univ., MBA, 2003*]
12. What are the major limitations of the PERT model? Discuss. [*Delhi Univ., MBA, 2004*]

SELF PRACTICE PROBLEMS A

1. Draw a network diagrams from the following list of activities:

Activity	*Predecessor Activity*			
	Set 1	*Set 2*	*Set 3*	*Set 4*
A	–	–	–	–
B	–	–	–	*A*
C	–	–	–	*A*
D	*A*	*A*	*A*	*B*
E	*B*	*A, B*	*A, B*	*B*
F	*B, C*	*A, B, C*	*B, C*	*D, E*
G	*D, E, F*	*D, E, F*	*C*	*D*
H	*E, F*	*F*	*D, E,F*	*C, F, G*

2. Following is the list of activities associated with the assembly of a space module for an upcoming mission. Draw the network diagram of activities involved in the project.

Activity	*Description*	*Predecessor Activity*
A	Construct shell of module	–
B	Order life support system and scientific experimentation package from supplier	–
C	Order components of control and navigation system	–
D	Wire module	*A*
E	Assemble control and navigational system	*C*
F	Preliminary test of life support system	*B*
G	Install life support system in module	*D, F*
H	Install scientific experimentation package in module	*D, F*
I	Preliminary test of control and navigational system	*E, F*
J	Install control and navigational system in module	*H, I*
K	Final testing and debugging	*G, J*

3. A new type of water pump is to be designed for an automobile. Its major specifications are given in the table below. Draw the network diagram of activates involved in the project.

Activity	*Description*	*Predecessor Activity*
A	Drawing prepared and approved	–
B	Cost analysis	*A*
C	Tool feasibility (economics)	*A*
D	Tool manufactured	*C*
E	Favourable cost	*B, C*
F	Raw materials procured	*D, E*
G	Subassemblies ordered	*E*
H	Subassemblies received	*G*
I	Parts manufactured	*D, F*
J	Final assembly	*I, H*
K	Testing and shipment	*J*

4. Listed in the table are the activities and sequencing required in the computerization of a bank branch.

Activity	*Description*	*Predecessor Activity*
A	Preparation/Deliberations in the department	–
B	Dialogue with the union	*A*
C	Discussion/Approval/Sanction of local management	*A*
D	Customer education	*B*
E	Preparing specifications for the system	*C*
F	Selection of staff	*B*
G	Order/Acquisition of system	*D, E*
H	Alterations in the branch premises	*C*
I	Training of staff	*F*
J	Wiring/Power-supply setup of the branch	*H*
K	Transition	*G, I, J*
L	Parallel run	*K*

Draw the network diagram of activities associated with the project.

5. The medical faculty of a university is considering to hold a faculty development programme. It has planned the following activities. Prepare a network diagram, showing the interrelationships of the various activities.

Activity	*Description*	*Predecessor*
A	Design conference meetings and theme	–
B	Design front cover of the conference proceedings	*A*
C	Prepare brochure and send request for papers	*A*
D	Compile list of distinguished speakers/guests	*A*
E	Finalize brochure and print it	*C, D*
F	Make travel arrangements for distinguished speakers/guests	*D*
G	Despatch brochures	*E*
H	Receive papers for conference	*G*
I	Edit papers and assemble proceedings	*F, H*
J	Print proceedings	*B, I*

[*Delhi Univ., MBA* (*HCA*), *2004*]

6. Activities and their description associated with the Haryali Watershed Project are listed below:

Activity	*Description*	*Predecessor Activity*
A	Consult General Water Board (GWB)	–
B	Motivate farmers	*A*
C	Identify beneficiaries spots	*B*
D	Obtain demand certificate from GWB	*C*
E	Obtain no objection certificate from GWB	*C*
F	Apply to DPAP	*C*
G	Forward to bank	*D, E, F*
H	Process application	*G*
I	Requisition drilling high	*G*
J	Release loans	*H*
K	Apply for power	*H*
L	Purchase pump sets	*I, J*
M	Drilling of walls	*I, J*
N	Construct storage tank	*I, J*
O	Prepare field channels	*I, J*
P	Test water for quality	*M*
Q	Install pump sets	*L, P*
R	Power connections	*K*
S	Tests	*R*

Draw the network diagram of activities associated with the project.

7. Listed in the table are the activities and sequencing requirement necessary for light system maintenance project of a particular stadium.

Activity	*Description*	*Predecessor Activity*
A	Assemble crew	–
B	Test lights for burned bulbs	–
C	Obtain needed bulbs	*B*
D	Paint light standards below banks	*A*
E	Replace burned bulbs	*C*
F	Deactivate system	*B*
G	Check all wiring for wear	*A, F*
H	Obtain needed wire	*G*
I	Clean lenses on lights	*A, F*
J	Remove worn wire	*G*
K	Cut new wire to needed lengths	*J, H*
L	Check insulators that support wires	*J*
M	Replace worn insulators	*L*
N	Replace old wire	*M, K*
O	Splice new wire with old	*N*
P	Insulate splices	*O*
Q	Paint light banks	*P*
R	Replace broken lenses	*E*
S	Reactivate system	*Q, D, I, R*
T	Clean up	*R*

Draw the network diagram of activities associated with the project.

8. Draw the network diagram of activities involved in the project of introduction of a new product.

Activity	*Description*	*Predecessor Activity*
A	Develop plan for introduction of the project	–
B	Prepare product drawings	*A*
C	Test and approve packages	*B*
D	Approve unit costs	*C*
E	Prepare and send questionnaires in the field	*D*
F	Evaluate questionnaires	*E*
G	Finalize labels, shipping boxes and cartons	*F*
H	Plan media advertising schedule and sales literature	*F*
I	Assess manpower needs, and hire and train employees	*F*
J	Design and develop production equipment	*B*
K	Arrange and instal production equipment	*J*
L	Debug equipment	*K*
M	Design and develop product and quality standards	*J*
N	Manufacture labels, shipping boxes and cartons	*G*
O	Order, receive and check raw materials	*I, L*
P	Prepare and check warehouse space	*F*
Q	Conduct trial run	*O*
R	Manufacture new product, first production run	*P, Q*
S	Conduct sales meetings	*H, I*
T	Contact customers and receive orders	*O, S*
U	Process orders and send the product	*R, T*

[*Delhi Univ., MBA, 2004*]

13.5 CRITICAL PATH ANALYSIS

The objective of critical path analysis is to estimate the total project duration and to assign starting and finishing times to all activities involved in the project. This helps to check the actual progress against the scheduled duration of the project.

The duration of individual activities may be uniquely determined (in case of CPM) or may involve the three time estimates (in case of PERT), out of which the expected duration of an activity is computed. Having done this, the following factors should be known in order to prepare the project scheduling.

(i) Total completion time of the project.
(ii) Earlier and latest start time of each activity.
(iii) Critical activities and critical path.
(iv) Float for each activity, i.e. the amount of time by which the completion of a non-critical activity can be delayed, without delaying the total project completion time.

Consider the following notations for the purpose of calculating various times of events and activities.

E_i = Earliest occurrence time of an event, i. This is the earliest time for an event to occur when all the preceding activities have been completed, without delaying the entire project.

L_i = Latest allowable time of an event, i. This is the latest time at which an event can occur without causing a delay in project's completion time.

ES_{ij} = Early starting time of an activity (i, j). This is the earliest time an activity should start without affecting the project completion.

Critical path is the longest path through the project network; the activities on the path are the critical activities, therefore any delay in their completion must be avoided to prevent delay in project completion.

LS_{ij} = Late starting time of an activity (i, j). This is the latest time an activity should start without delaying the project completion.

EF_{ij} = Early finishing time of an activity (i, j). This is the earliest time an activity should finish without affecting the project completion.

LF_{ij} = Late finishing time of an activity (i, j). This is the latest time an activity should finish without delaying the project completion.

t_{ij} = Duration of an activity (i, j).

As mentioned earlier, a network diagram should have only one initial event and one end event. The other events are numbered consecutively with integer 1, 2, . . ., n, such that $i < j$ for any two events i and j connected by an activity, which starts at i and finishes at j.

For calculating the earliest occurrence and latest allowable times for events, following two methods: *Forward Pass* method and *Backward Pass* method are used:

13.5.1 Forward Pass Method (For Earliest Event Time)

In this method, calculations begin from the initial event 1, proceed through the events in an increasing order of event numbers and end at the final event, say N. At each event, its *earliest occurrence time* (E) and earliest start and finish time for each activity that begins at that event is calculated. When calculations end at the final event N, its earliest occurrence time gives the earliest possible completion time of the project. The method may be summarized as follows:

1. Set the earliest occurrence time of initial event 1 to zero. That is, $E_1 = 0$, for $i = 1$.
2. Calculate the earliest start time for each activity that begins at event i (= 1). This is equal to the earliest occurrence time of event, i (tail event). That is:
$$ES_{ij} = E_i, \quad \text{for all activities } (i, j) \text{ starting at event } i.$$
3. Calculate the earliest finish time of each activity that begins at event i. This is equal to the earliest start time of the activity plus the duration of the activity. That is:
$$EF_{ij} = ES_{ij} + t_{ij} = E_i + t_{ij}, \quad \text{for all activities } (i, j) \text{ beginning at event } i.$$
4. Proceed to the next event, say j; $j > i$.
5. Calculate the earliest occurrence time for the event j. This is the maximum of the earliest finish times of all activities ending into that event, that is,
$$E_j = \text{Max } \{EF_{ij}\} = \text{Max } \{E_i + t_{ij}\}, \quad \text{for all immediate predecessor activities.}$$
6. If $j = N$ (final event), then earliest finish time for the project, that is, the earliest occurrence time E_N for the final event is given by
$$E_N = \text{Max } \{EF_{ij}\} = \text{Max } \{E_{N-1} + t_{ij}\}, \quad \text{for all terminal activities}$$

13.5.2 Backward Pass Method (For Latest Allowable Event Time)

In this method, calculations begin from the final event N. Proceed through the events in the decreasing order of event numbers and end at the initial event 1. At each event, *latest occurrence time* (L) and latest finish and start time for each activity that is terminating at that event is calculated. The procedure continues till the initial event. The method may be summarized as follows:

1. Set the latest occurrence time of last event, N equal to its earliest occurrence time (known from forward pass method). That is, $L_N = E_N, \quad j = N$.
2. Calculate the latest finish time of each activity which ends at event j. This is equal to latest occurrence time of final event. That is:
$$LF_{ij} = L_i, \quad \text{for all activities } (i, j) \text{ ending at event } j.$$
3. Calculate the latest start times of all activities ending at j. This is obtained by subtracting the duration of the activity from the latest finish time of the activity. That is:
$$LF_{ij} = L_j$$
and $$LS_{ij} = LF_{ij} - t_{ij} = L_j - t_{ij}, \quad \text{for all activity } (i, j) \text{ ending at event } j.$$
4. Proceed backward to the event in the sequence, that decreases j by 1.
5. Calculate the latest occurrence time of event i ($i < j$). This is the minimum of the latest start times of all activities from the event. That is:
$$L_i = \text{Min } \{LS_{ij}\} = \text{Min } \{L_j - t_{ij}\}, \text{ for all immediate successor activities.}$$
6. If $j = 1$ (initial event), then the latest finish time for project, i.e. latest occurrence time L_1 for the initial event is given by:
$$L_1 = \text{Min } \{LS_{ij}\} = \text{Min}\{L_{j-1} - t_{ij}\}, \text{ for all immediate successor activities.}$$

13.5.3 Float (Slack) of an Activity and Event

The *float* (*slack*) or *free time* is the length of time in which a non-critical activity and/or an event can be delayed or extended without delaying the total project completion time.

Slack of an Event

The *slack* (or *float*) of an event is the difference between its latest occurrence time (L_i) and its earliest occurrence time (E_i). That is:

$$\text{Event float} = L_i - E_i$$

It is a measure of how long an event can be delayed without increasing the project completion time.

(a) If $L = E$ for certain events, then such events are called *critical events*.

(b) If $L \neq E$ for certain events, then the float (slack) on these events can be negative ($L < E$) or positive ($L > E$).

Slack of an Activity

It is the amount of activity time that can increased or delayed without delaying project completion time. This float is calculated as the difference between the latest finish time and the earliest finish time for the activity. There are three types of floats for each non-critical activity in a project.

***(a) Total float*:** This is the length of time by which an activity can be delayed until all preceding activities are completed at their earliest possible time and all successor activities can be delayed until their latest permissible time.

For each non-critical activity (i, j) the total float is equal to the latest allowable time for the event at the end of the activity *minus* the earliest time for an event at the begining of the activity *minus* the activity duration. That is:

$$\text{Total float } (TF_{ij}) = (L_j - E_i) - t_{ij} = LS_{ij} - ES_{ij} = LF_{ij} - EF_{ij}$$

***(b) Free float*:** This is the length of time by which the completion time of any non-critical activity can be delayed without causing any delay in its immediate successor activities. The amount of free float time for a non-critical activity (i, j) is computed as follows:

$$\begin{aligned}\text{Free float } (FF_{ij}) &= (E_j - E_i) - t_{ij} \\ &= \text{Min } \{ES_{ij}, \text{ for all immediate successors of activity } (i, j)\} - EF_{ij}\end{aligned}$$

***(c) Independent float*:** This is the length of time by which completion time of any non-critical activity (i, j) can be delayed without causing any delay in its predecessor or the successor activities. Independent float time for each non-critical activity is computed as follows:

$$\text{Independent float } (IF_{ij}) = (E_j - L_i) - t_{ij} = \{ES_{ij} - LS_{ij}\} - t_{ij}$$

The negative value of independent float is considered to be zero.

Remarks 1. Latest occurrence time of an event is always greater than or equal to its earliest occurrence time (i.e. $L_i \geq E_i$),

$$TF_{ij} \geq (L_j - E_i) - t_{ij}$$

This implies that the value of free float may range from zero to total float but will not exceed total float value. That is, Independent float ≤ Free float ≤ Total float.

2. The calculation of various floats can help the decision-maker in identifying the underutilized resources, flexibility in the total schedule and possibilities of redeployment of resources.
3. Total float for a non-critical activity may be viewed as follows:
   (a) ***Negative (i.e. L – E < 0):*** Project completion is behind the schedules date, i.e., resources are not adequate and activities may not finish in time. This needs extra resources or certain activities need crashing in order to reduce negative float value.
   (b) ***Positive (i.e. L – E > 0):*** Project completion is ahead of the schedule date, i.e., resources are surplus. These resources can be deployed elsewhere or execution of the activities can be delayed.
   (c) ***Zero (i.e. L = E):*** Resources are just sufficient for the completion of activities in a project. Any delay in activities execution will necessarily increase the project cost and time.

13.5.4 Critical Path

Certain activities in any project are called *critical activities* because delay in their execution will cause further delay in the project completion time. All activities having zero total float value are identified as critical activities, i.e., $L = E$

The *critical path* is the sequence of critical activities between the start event and end event of a project. This is critical in the sense that if execution of any activity of this sequence is delayed, then completion of the project will be delayed. A critical path is shown by a thick line or double lines in the network diagram.

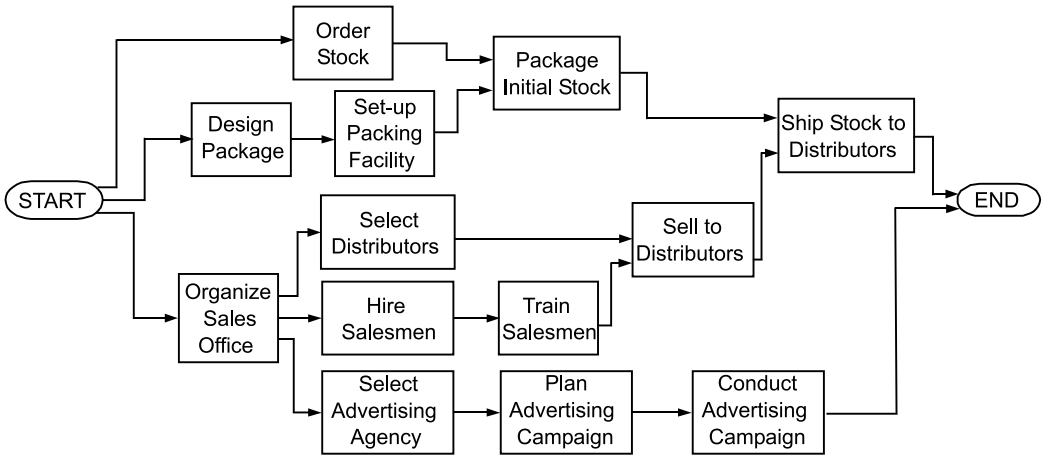
The *length of the critical path* is the sum of the individual completion times of all the critical activities and defines the longest time to complete the project. The critical path in a network diagram can be identified as:

(i) If E_i-value and L_j-value for any tail and head events is equal, then activity (i, j) between such events is referred as critical, That is, $E_j = L_j$ and $E_i = L_i$.
(ii) On critical path $E_j - E_i = L_j - L_i = t_{ij}$.

Example 13.4 An established company has decided to add a new product to its line. It will buy the product from a manufacturing concern, package it, and sell it to a number of distributors that have been selected on a geographical basis. Market research has already indicated the volume expected and the size of sales force required. The steps shown in the following table are to be planned.

Activity	*Description*	*Predecessors*	*Duration (days)*
A	Organize sales office	–	6
B	Hire salesmen	*A*	4
C	Train salesmen	*B*	7
D	Select advertising agency	*A*	2
E	Plan advertising campaign	*D*	4
F	Conduct advertising campaign	*E*	10
G	Design package	–	2
H	Setup packaging facilities	*G*	10
I	Package initial stocks	*J*, *H*	6
J	Order stock from manufacturer	–	13
K	Select distributors	*A*	9
L	Sell to distributors	*C*, *K*	3
M	Ship stocks to distributors	*I*, *L*	5

The precedence relationship among these activities are shown in the following figure.



As the figure shows, the company can begin to organize the sales office, design the package, and order the stock immediately. Also the stock must be ordered and the packing facility must be set up before the initial stocks are packaged.

(a) Draw an arrow diagram for this project.
(b) Indicate the critical path.
(c) For each non-critical activity, find the total and free float. [*Delhi Univ., MBA, Dec. 2005*]

Solution (a) The arrow diagram for the given project, along with E-values and L-values, is shown in Fig. 13.11. Determine the earliest start time – E_i and the latest finish time – L_j for each event by proceeding as follows:

Forward Pass Method

$E_1 = 0$

$E_3 = E_1 + t_{1,3} = 0 + 2 = 2$

$E_5 = E_2 + t_{2,5} = 6 + 4 = 10$

$E_6 = \underset{i=2,5}{\text{Max}} \{E_i + t_{i,6}\} = \text{Max} \{E_2 + t_{2,6}; E_5 + t_{5,6}\}$

$= \text{Max} \{6 + 9; 10 + 7\} = 17$

$E_7 = E_2 + t_{2,7} = 6 + 2 = 8$

$E_9 = \underset{i=4,6}{\text{Max}} \{E_i + t_{i,9}\} = \text{Max} \{E_4 + t_{4,9}; E_6 + t_{6,9}\}$

$= \text{Max} \{13 + 6; 17 + 3\} = 20$

$E_2 = E_1 + t_{1,2} = 0 + 6 = 6$

$E_4 = \underset{i=1,3}{\text{Max}} \{E_i + t_{i,4}\}$

$= \text{Max} \{E_1 + t_{14}; E_3 + t_{34}\}$

$= \text{Max} \{0 + 13, 2 + 10\} = 13$

$E_8 = E_7 + t_{7,8} = 8 + 4 = 12$

$E_{10} = \underset{i=8,9}{\text{Max}} \{E_i + t_{i,10}\}$

$= \text{Max} \{E_8 + t_{8,10}; E_9 + t_{9,10}\}$

$= \text{Max} \{12 + 10; 20 + 5\} = 25.$

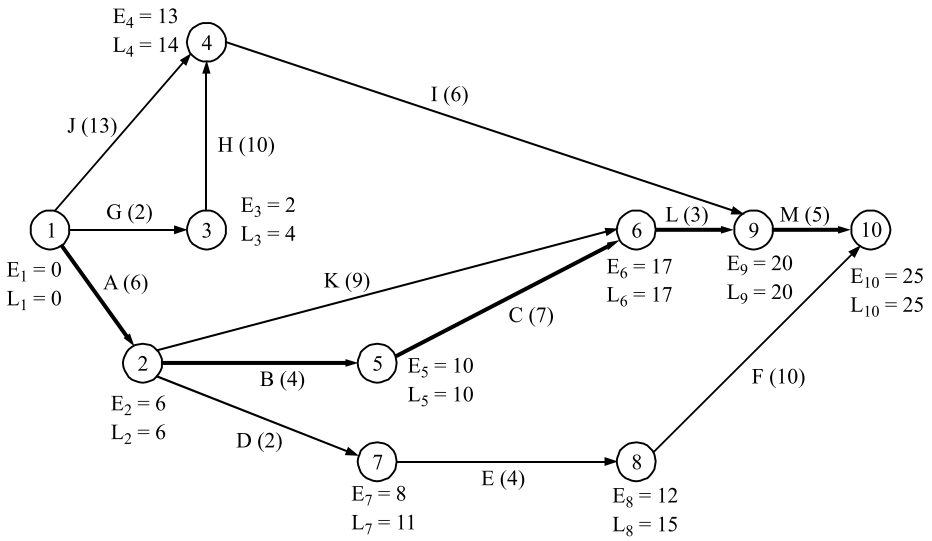


Fig. 13.11
Network Diagram

Backward Pass Method

$L_{10} = E_{10} = 25$

$L_8 = L_{10} - t_{8,10} = 25 - 10 = 15$

$L_6 = L_9 - t_{6,9} = 20 - 3 = 17$

$L_4 = L_9 - t_{4,9} = 20 - 6 = 14$

$L_2 = \underset{j=5,6,7}{\text{Min}} \{L_j - t_{2,j}\}$

$= \text{Min} \{L_5 - t_{2,5};\ L_6 - t_{2,6};\ L_7 - t_{2,7}\}$

$= \text{Min} \{10 - 4;\ 17 - 9;\ 11 - 2\} = 6$

$L_9 = L_{10} - t_{9,10} = 25 - 5 = 20$

$L_7 = L_8 - t_{7,8} = 15 - 4 = 11$

$L_5 = L_6 - t_{5,6} = 17 - 7 = 10$

$L_3 = L_4 - t_{3,4} = 14 - 10 = 4$

$L_1 = \underset{j=2,3,4}{\text{Min}} \{L_j - t_{1,j}\}$

$= \text{Min} \{L_2 - t_{1,2};\ L_3 - t_{1,3};\ L_4 - t_{1,4}\}$

$= \text{Min} \{6 - 6;\ 4 - 2;\ 14 - 13\} = 0$

(b) The critical path in the network diagram (Fig. 13.11) has been shown. This has been done by double lines by joining all those events where E-values and L-values are equal.

The critical path of the project is: 1 – 2 – 5 – 6 – 9 – 10 and critical activities are *A, B, C, L* and *M*. The total project completion time is 25 weeks.

(c) For each non-critical activity, the total float and free float calculations are shown in Table 13.1.

Table 13.1 Total Float and Free Float

Activity (i, j)	*Duration* (t_{ij})	*Earliest Time*		*Latest Time*		*Float*	
		Start (E_i)	*Finish* $(E_i + t_{ij})$	*Start* $(L_j - t_{ij})$	*Finish* L_j	*Total* $(L_j - t_{ij}) - E_i$	*Free* $(E_j - E_i) - t_{ij}$
1 – 3	2	0	2	2	4	2	0
1 – 4	13	0	13	1	14	1	0
2 – 6	9	6	15	8	17	2	2
2 – 7	2	6	8	9	11	3	0
3 – 4	10	2	12	4	14	2	1
4 – 9	6	13	19	14	20	1	1
7 – 8	4	8	12	11	15	3	0
8 – 10	10	12	22	15	25	3	3

Example 13.5 An insurance company has decided to modernize and refit one of its branch offices. Some of the existing office equipments will be disposed of but the remaining will be returned to the branch after the completion of the renovation work. Tenders are invited from a number of selected contractors. The contractors would be responsible for all the activities in connection with the renovation work excepting the prior removal of the old equipment and its subsequent replacement.

The major elements of the project have been identified, as follows, along with their durations and immediately preceding elements.

Activity	*Description*	*Duration (weeks)*	*Immediate Predecessors*
A	Design new premises	14	–
B	Obtain tenders from the contractors	4	*A*
C	Select the contractor	2	*B*
D	Arrange details with selected contractor	1	*C*
E	Decide which equipment is to be used	2	*A*
F	Arrange storage of equipment	3	*E*
G	Arrange disposal of other equipment	2	*E*
H	Order new equipment	4	*E*
I	Take delivery of new equipment	3	*H, L*
J	Renovations take place	12	*K*
K	Remove old equipment for storage or disposal	4	*D, F, G*
L	Cleaning after the contractor has finished	2	*J*
M	Return old equipment for storage	2	*H, L*

(a) Draw the network diagram showing the interrelations between the various activities of the project.
(b) Calculate the minimum time that the renovation can take from the design stage.

(c) Find the effect on the overall duration of the project if the estimates or tenders can be obtained in two weeks from the contractors by reducing their numbers.

(d) Calculate the 'independent float' that is associated with the non-critical activities in the network diagram.

Solution (a) The network diagram for the given project, along with *E*-values and *L*-values, is shown in Fig. 13.12.

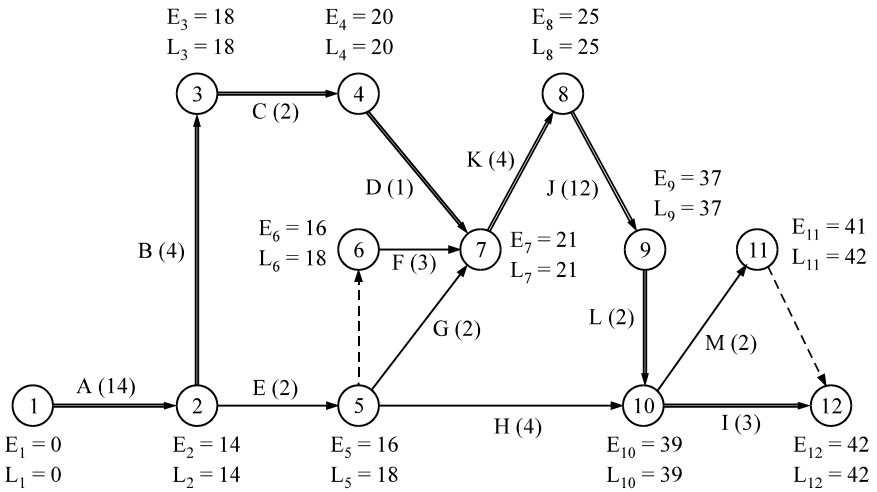


Fig. 13.12 Network Diagram

The critical path in the network diagram (Fig 13.12) has been shown by double lines joining all those events where *E*-values and *L*-values are equal.

(b) The critical path of the project is: 1 – 2 – 3 – 4 – 7 – 8 – 9 – 10 – 12 and critical activities are *A, B, C, D, K, J, L* and *I*. The total project completion time is 42 weeks.

For non-critical activities, the total float, free float and independent float calculations are shown in Table 13.2.

Activity	*Duration*	*Earliest Time*		*Latest Time*		*Float*		
(i, j)	(t_{ij})	*Start* (E_i)	*Finish* $(E_i + t_{ij})$	*Start* $(L_j - t_{ij})$	*Finish* L_j	*Total* $(L_j - t_{ij}) - E_i$	*Free* $(E_j - E_i) - t_{ij}$	*Independent* $(E_j - L_i) - t_{ij}$
2 – 5	2	14	16	16	18	2	0	0
6 – 7	3	16	19	18	21	2	2	0
5 – 7	2	16	18	19	21	3	3	1
5 – 10	4	16	20	35	39	19	19	17
10 – 11	2	39	41	40	42	1	0	0

Table 13.2 Calculation of Floats

(c) The effect on the overall project duration, if the time of activity B is reduced to 2 weeks instead of 4 weeks, is shown in Table 13.3.

Path	*Duration*	
(i) A – E – H – I	23	
(ii) A – E – H – M	22	
(iii) A – B – C – D – K – J – L – I (Critical path 42 weeks)	40	(New critical path)
(iv) A – B – C – D – K – J – L – M (Critical path 41 weeks)	39	
(v) A – E – G – K – J – L – I	39	
(vi) A – E – G – K – J – L – M	39	
(vii) A – E – F – K – J – L – I	40	(New critical path)
(viii) A – E – F – K – J – L – M	39	

Table 13.3

CONCEPTUAL QUESTIONS B

1. Explain the following terms in PERT/CPM.
   (i) Earliest time, (ii) Latest time, (iii) Total activity time, (iv) Event slack, and (v) Critical path.
2. (a) What is float? What are the different types of floats?
   (b) Discuss in brief: (i) total float, and (ii) free float. Also explain their uses in network.
3. What is meant by the term critical activities, and why is it necessary to know about them?
4. Explain the significance of 'working out of float' in the network of project activities. Discuss, in brief, the different types of floats. [*Delhi Univ., MBA, 1998*]
5. Explain the following terms in the context for project management:
   (i) Resources float, (ii) Activity variance, (iii) Project variance.

SELF PRACTICE PROBLEMS B

1. An architect has been awarded a contract to prepare plans for an urban renewal project. The job consists of the following activities and their estimated times:

Activity	*Description*	*Immediate Predecessors*	*Time (days)*
A	Prepare preliminary sketches	–	2
B	Outline specifications	–	1
C	Prepare drawings	*A*	3
D	Write specifications	*A, B*	2
E	Run off prints	*C, D*	1
F	Have specification	*B, D*	3
G	Assemble bid packages	*E, F*	1

   (a) Draw the network diagram of activities for the project.
   (b) Indicate the critical path, and calculate the total float and free float for each activity.

2. A research and development department is developing a new power supply for a console television set. It has broken the job down into the following:

Job	*Description*	*Immediate Predecessors*	*Time (days)*
A	Determine output voltages	–	5
B	Determine whether to use solid state rectifiers	*A*	7
C	Choose rectifier	*B*	2
D	Choose filters	*B*	3
E	Choose transformer	*C*	1
F	Choose chassis	*D*	2
G	Choose rectifier mounting	*C*	1
H	Layout chassis	*E, F*	3
I	Build and test	*G, H*	10

   (a) Draw the network diagram of activities involved in the project and indicate the critical path.
   (b) What is the minimum completion time for the project?

3. The following maintenance job has to be performed periodically on the heat exchangers in a refinery:

Task	*Description*	*Immediate Predecessors*	*Time (days)*
A	Dismantle pipe connections	–	14
B	Dismantle header, closure, and floating head front	*A*	22
C	Remove tube bundle	*B*	10
D	Clean bolts	*B*	16
E	Clean header and floating head front	*B*	12
F	Clean tube bundle	*C*	10
G	Clean shell	*C*	6
H	Replace tube bundle	*F, G*	8
I	Prepare shell pressure test	*D, E, H*	24
J	Prepare tube pressure test and make the final reassembly	*I*	16

   (a) Draw a network diagram of activities for the project.
   (b) Identify the critical path. What is its length?
   (c) Find the total float and free float for each task. [*Delhi Univ., MBA, 2004*]

4. The sales manager of Domestic Products Limited was informed by the company's R&D department about the completion of the prototype of a particular product. He consulted the production manager on the time taken to produce the first batch of the product, which is needed for demonstration in his sales promotion programme. He also decided to invite a few industrial representatives to the demonstration of this new product and through them to launch it in the market. The various activities involved in this marketing project, their descriptions, estimated duration (in days) and immediate predecessors are given in the following table:

Activity	*Description*	*Time (days)*	*Immediate Predecessors*
A	Collect data on specifications and capabilities	4	–
B	Prepare operation manual	4	*A*
C	Chart out promotion programme	4	*B*
D	Make copies of manual and promotion material	9	*B*
E	Produce first batch for demonstration	16	*B*
F	Prepare list of press representatives	2	*C*
G	Chief executive's conference with Managers	1	*C*
H	Press representatives reach Bombay	2	*F, G*
I	Promotional meetings	4	*D, H*
J	Product demonstration	2	*E, I*
K	Press representatives return home	2	*J*

   (a) Draw the network diagram for the project.
   (b) Identify the critical path. What is the maximum time required to complete the project.
   (c) Find the total float and free float (if any) for all the non-critical activities.

5. Listed in the table are the activities and sequencing requirements necessary for the completion of a research report.

Activity	Description	Time (days)	Immediate Predecessors
A	Literature search	–	6
B	Formulation of hypothesis	–	5
C	Preliminary feasibility study	B	2
D	Formal proposal	C	2
E	Field analysis	A, D	2
F	Progress report	D	1
G	Formal research	A, D	6
H	Data collection	E	5
I	Data analysis	G, H	6
J	Conclusions	I	2
K	Rough draft	G	4
L	Final copy	J, K	3
M	Preparation of oral presentation	L	1

(a) Draw a network diagram for the project.
(b) Find the critical path. What is its length?
(c) Find the total float and the free float for each non-critical activity.

6. Listed in the table are the activities and their simplified sequencing requirements for publishing a textbook.

Activity	Description	Preceding Activity	Expected Completion Time (months)
A	Write book	–	12
B	Design book	A	1
C	Edit manuscript	A	6
D	Check editing	C	2
E	Accept design	B	1
F	Copy edit	D, E	2
G	Prepare artwork	D, E	4
H	Accept and correct artwork	G	1/2
I	Set galleys	F	4
J	Check and correct galleys	I	1
K	Full page proofs	H, J	2
L	Check and correct pages	K	1
M	Prepare index	K	1
N	Set and correct index	M	1/2
O	Check camera-ready copy	L, N	1/2
P	Print and bind book	O	1

(a) Construct the network of activities and find the critical path.
(b) For each non-critical activity, find total the float and free float.

7. Delhi Medical Association is considering to hold a conference. The following table gives the list of activities involved, their immediate predecessors, and their duration (in days):

Activity	Description	Predecessor	Duration (days)
A	Design conference meetings and theme	–	3
B	Design front cover of the conference proceedings	A	2
C	Prepare brochure and send request for papers	A	6
D	Compile list of distinguished speakers/guests	A	3
E	Finalize brochure and print it	C, D	7
F	Make travel arrangements for speakers/guests	D	4
G	Despatch brochures	E	3
H	Receive papers for conference	G	25
I	Edit papers and assemble proceedings	F, H	10
J	Print proceedings	B, I	20

(a) Prepare a network diagram showing the interrelationships of the various activities.
(b) Find the total time required to hold the conference.
(c) Compute the total float for the non-critical activities.
[*Delhi Univ., MBA (HCA), 2002*]

8. The following table gives a list of various activities and their immediate predecessors involved in installation of CAT scanner in a hospital:

Activity	Description	Immediate Predecessors	Expected Duration (days)
A	Finalization of the layout plans	–	2
B	Demolition of the structures	A	6
C	Walls erection	B	12
D	Flooring	B	8
E	Electrical wiring	C	6
F	Air conditioning ducting	C	4
G	Fire alarm installation	C	3
H	False ceiling and light fittings	E, F, G	10
I	Wall plastering and painting	H	9
J	Equipment installation	D, I	6
K	Calibration and testing	J	3
L	Final finishing	K	2
M	Handing over	L	1

(a) Prepare a network diagram for the project.
(b) Find the total time required to install CAT scanner.
(c) Calculate the total float for various non-critical activities.
[*Delhi Univ., MBA (HCA), 2000, 2003*]

9. A medical association prepares an annual programme each year, giving the monthly meeting dates, background on the speakers, an abstract of their talks, and an alphabetic listing, both by name and medical college/hospital affiliation, of all dues paying members. The programme is mailed to these members as well as to selected individuals and organizations. The activities to be performed are listed as follows:

Activity	Description	Preceding Activities	Estimated Time (weeks)
A	Decide on general orientation for this year's programme	–	1
B	Get commitments from speakers and abstracts of their talks	A	4
C	Solicit advertising to appear in the programme	A	3
D	Mail out dues notices and wait for response	–	6
E	Prepare list of dues-paying members	D	1
F	Give copy to printer before proofreadng	B, C, E	2
G	Get programme printed and assembled	F	2
H	Prepare final mailing list	E	1
I	Staff envelopes and mail programmes	G, H	1

(a) Develop a network diagram to show the relationship between all activities. Specify the activities on the critical path and the project completion time.

(b) The programme chairman now claims that it will take him 6 weeks to get commitments and abstracts from the speakers. Will this delay the project completion time? How will it affect the critical path?

[*Delhi Univ., MBA (HCA), 2005*]

10. A reactor and storage tank are interconnected by a 3″ insulated process line that needs periodic replacement. There are valves along the lines and at the terminals and these need replacing as well. No pipe and valves are in stock. Accurate, as built, drawings exist and are available. This line is overhead and requires scaffolding. Pipe sections can be shop-fabricated at the plant. Adequate craft labour is available.

You are the maintenance and construction superintendent responsible for his project. The works engineer has requested your plan and schedule for a review with the operating supervision. The plant methods and standards section has furnished the following data. The precedent for each activity have been determined from a familiarity with similar projects.

Activity	*Description*	*Time (hrs)*	*Precedents*
A	Develop required material list	8	–
B	Procure pipe	200	*A*
C	Erect scaffold	12	–
D	Remove scaffold	4	*H, M*
E	Deactivate line	8	–
F	Prefabricate sections	40	*B*
G	Place new pipes	32	*F, K*
H	Fit up pipe and valves	8	*G, J*
I	Procure valves	225	*A*
J	Place valves	8	*I, K*
K	Remove old pipe and valves	35	*C, E*
L	Insulate	24	*G, J*
M	Pressure test	6	*H*
N	Clean-up and start-up	4	*D, M*

(a) Sketch the network diagram of the project plan.
(b) Make the forward pass and backward pass calculations on this network, and indicate the critical path and its length.
(c) Calculate the total float and free float for each of the non-critical activities.

11. A company has decided to market a new product for the consumer market. The problems of how to plan and control the various phases of this project – sales promotion, training of salespeople, pricing, packaging, advertising, and manufacturing – are obvious to the management of this firm. They have asked you to guide them through this difficult venture using CPM. The first firm to market this type of product will reap substantial profits and will enhance its image by marketing such a revolutionary product. A list of the activities, with the expected time duration for each, is given in the table, in terms of weeks.

Activity	*Description*	*Precedence*	*Time (weeks)*
Manufacturing activities			
A	Study equipment requirement	–	1
B	Select supplier of equipment	*A*	1
C	Study manufacturing procedures	*B*	3
D	Study quality control procedures	*C*	3
E	Study purchasing and inventory rules	*B*	3
F	Receive and install equipment	*B*	8
G	Place order for raw materials	*E*	2
H	Manufacture from raw materials for test and first production runs	*G*	4
I	Receive containers and packaging supplies	*P*	1
J	Have personnel available for first production run	*F*	0
K	Run manufacturing test	*D, J, H, I, T*	3
L	Run first production	*K*	7
Marketing activities			
M	Price product	*B, S*	2
N	Prepare artwork for advertising	*M*	4
O	Send out advertising materials and packaging orders to suppliers	*N*	5
P	Produce advertising and packaging materials	*O*	1
Q	Hold sales meeting	*K, S*	2
R	Training salespeople	*Q*	3
Accounting activities			
S	Determine cost of new product	*B*	2
T	Determine cost of the new product inventory	*S*	3

(a) Draw a network diagram for the project.
(b) Identify the critical path and determine its length.

12. In Border Roads Organization, the vehicles and equipment are being overhauled in base workshop. The overhauling of vehicles involves a number of jobs. These are given in the following table.

Job	*Description*	*Immediate Predecessors*	*Expected Time (days)*
A	Dismantling of vehicle	–	3
B	Inspection and preparation of visual inspection on report	*A*	2
C	Stripping minor assys	*B*	2
D	Stripping major assys	*B*	3
E	Overhauling engine	*C*	40
F	Cleaning chassis	*C, D*	2
G	Cleaning body and cabin	*C, D*	2
H	Repairing gear box	*C*	2
I	Repairing break system	*D*	1
J	Repairing axle and propeller shaft	*D*	6
K	Repairing road spring	*D*	2
L	Repairing chassis	*F*	2
M	Repairing body and cabin	*G*	5
N	Placing chassis and fitment of engine	*E, L*	2
O	Fitting major assys	*N, H*	1
P	Fitting minor assys and cabin	*N, I, J, K, M*	2
Q	Checking and rectification	*O, P*	1
R	Mechanical inspection for passing	*Q*	1
S	Painting	*R*	1
T	Checking and rectification before final inspection	*S*	1
U	Final inspection and passing	*T*	1
V	Closing the job card	*U*	1

(a) Draw a network diagram for the overhauling project.
(b) Identify the critical path. What is its length?
(c) Find the total float and free float for each task.

HINTS AND ANSWERS

1. A – D – F – G
2. A – B – D – F – H – I; 30 days
3. A – B – C – F – H – I – J; 104 days
4. A – B – E – J – K; 28 days
5. B – C – D – E – H – I – J – L – M; 28 days
9. A – B – C – E – N – P – Q – R – S – T – U – V; 56 days

13.6 PROJECT SCHEDULING WITH UNCERTAIN ACTIVITY TIMES

PERT was developed to handle projects where the time duration for each activity is not known with certainty but is a random variable that is characterized by β (beta)-distribution. To estimate the parameters: *mean and variance*, of the β-distribution three time estimates for each activity are required to calculate its expected completion time. The three-time estimates that are required are as under.

(i) ***Optimistic time*** (t_o ***or a***) The shortest possible time (duration) in which an activity can be performed assuming that everything goes well.

(ii) ***Pessimistic time*** (t_p ***or b***) The longest possible time required to perform an activity under extremely bad conditions. However, such conditions do not include natural calamities like earthquakes, flood, etc.

(iii) ***Most likely time*** (t_m ***or m***) The time that would occur most often to complete an activity, if the activity was repeated under exactly the same conditions many times. Obviously, it is the completion time that would occur most frequently (i.e. model value).

The β-distribution is not necessarily symmetric, the degree of skewness depends on the location of t_m to t_o and t_p. The range of optimistic time (t_o) and pessimistic time (t_p) is assumed to enclose every possible duration of the activity. The most likely completion time (t_m) for an activity may not be equal to the midpoint $(t_o + t_p)/2$ and may occur to its left or to its right as shown in Fig. 13.13.

In Beta-distribution the midpoint $(t_o + t_p)/2$ is given half weightage than that of most likely point (t_m). Thus, the expected or mean (t_e or μ) time of an activity, that is also the *weighted average* of three time estimates, is computed as the arithmetic mean of $(t_o + t_p)/2$ and 2 t_m. That is:

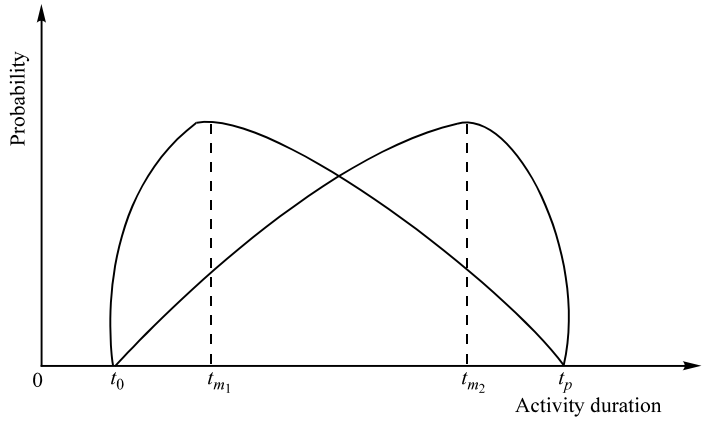


Fig. 13.13 An Example of Beta Distribution

$$\text{Expected time of an activity}\,(t_e) = \frac{(t_o + t_p)/2 + 2\,t_m}{3} = \frac{t_o + 4\,t_m + t_p}{6}$$

If duration of activities associated with the project is uncertain, then *variance* describes the dispersion (variation) in the activity time values. The calculations are based on the concept of normal distribution where 99 per cent of the area under normal curve falls within ± 3σ from the mean or fall within the range approximately 6 standard deviation in length. Therefore, the interval (t_o, t_p) or range ($t_p - t_o$) is assumed to enclose about 6 standard deviations of a symmetric distribution. Thus, if σ_i is the standard deviation of the duration of activity *i*, then

$$6\,\sigma_i \cong t_p - t_o \quad \text{or} \quad \sigma_i = \frac{t_p - t_o}{6}$$

$$\text{Variance of activity time, } \sigma_i^2 = \left[\frac{1}{6}(t_p - t_o)\right]^2$$

If duration of activities is a random variable, then the variance of the total *critical path's* duration is obtained by adding variances of individual critical activities. Suppose σ_c is the standard deviation of the critical path. Then:

$$\sigma_c^2 = \sum \sigma_i^2 \quad \text{and} \quad \sigma_c = \sqrt{\sum \sigma_i^2}$$

13.6.1 Estimation of Project Completion Time

Due to variation in the activity completion time, there are chances of variation in the scheduled completion time of the project. Thus, decision-maker needs to understand the probability of actually meeting the scheduled time.

The probability distribution of times for completing an event can be approximated by the normal distribution using statement of central limit theorem. Thus, the probability of completing the project on the schedule (or desired) time, T_s is given by:

$$\text{Prob}\left(Z = \frac{T_s - T_e}{\sigma_i}\right)$$

where, T_e = expected completion time of the project

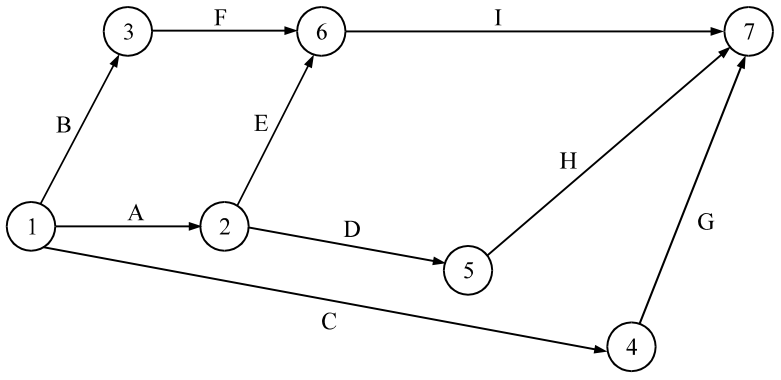
Z = number of standard deviations, the scheduled completion time is away from the expected (mean) time.

$\sigma_i^2 = \sigma_1^2 + \sigma_2^2 + \ldots + \sigma_n^2$ is the sum of variances of critical activities.

Hence, revised scheduled completion time of the project can be calculated as: $T_s = Z\,\sigma_i + T_e$, where value of Z corresponds to the probability of project completion time. The computation of T_s enables a decision-maker to make certain commitments, knowing the degree of risk. The expected completion time (T_e) of the project is obtained by adding the expected time of each critical activity.

Example 13.6 The following network diagram represents activities associated with a project:

Activities	:	A	B	C	D	E	F	G	H	I
Optimistic time, t_0	:	5	18	26	16	15	6	7	7	3
Pessimistic time, t_p	:	10	22	40	20	25	12	12	9	5
Most likely time, t_m	:	8	20	33	18	20	9	10	8	4



Determine the following:

(a) Expected completion time and variance of each activity

(b) The earliest and latest expected completion times of each event.

(c) The critical path.

(d) The probability of expected completion time of the project if the original scheduled time of completing the project is 41.5 weeks.

(e) The duration of the project that will have 95 per cent chance of being completed.

Solution (a) Calculations for expected completion time (t_e) of an activity and variance (σ^2), using following formulae are shown in Table 13.4.

$$t_e = \frac{1}{6}(t_o + 4t_m + t_p) \quad \text{and} \quad \sigma_i^2 = \left\{\frac{1}{6}(t_p - t_o)\right\}^2$$

(b) The earliest and latest expected completion time for all events considering the expected completion time of each activity are shown in Table 13.4.

Activity	t_o	t_p	t_m	$t_e = \frac{1}{6}(t_o + 4t_m + t_p)$	$\sigma^2 = [\frac{1}{6}(t_p - t_o)]^2$
1 – 2	5	10	8	7.8	0.696
1 – 3	18	22	20	20.0	0.444
1 – 4	26	40	33	33.0	5.429
2 – 5	16	20	18	18.0	0.443
2 – 6	15	25	20	20.0	2.780
3 – 6	6	12	9	9.0	1.000
4 – 7	7	12	10	9.8	0.694
5 – 7	7	9	8	8.0	0.111
6 – 7	3	5	4	4.0	0.111

Table 13.4

Forward Pass Method

$E_1 = 0$
$E_2 = E_1 + t_{1,2} = 0 + 7.8 = 7.8$
$E_3 = E_1 + t_{1,3} = 0 + 20 = 20$
$E_4 = E_1 + t_{1,4} = 0 + 33 = 33$
$E_5 = E_2 + t_{2,5} = 7.8 + 18 = 25.8$
$E_6 = \text{Max } \{E_i + t_{i,6}\} = \text{Max } \{E_2 + t_{2,6}\,;\, E_3 + t_{3,6}\}$
$= \text{Max } \{7.8 + 20;\ 20 + 9\} = 29$
$E_7 = \text{Max } \{E_i + t_{i,7}\}$
$= \text{Max } \{E_4 + t_{4,7}\,;\, E_5 + t_{5,7}\,;\, E_6 + t_{6,7}\}$
$= \text{Max } \{33 + 9.8\,;\, 25.8 + 8;\ 29 + 4\} = 42.8$

Backward Pass Method

$L_7 = E_7 = 42.8$
$L_6 = L_7 - t_{6,7} = 42.8 - 4 = 38.8$
$L_5 = L_7 - t_{5,7} = 42.8 - 8 = 34.8$
$L_4 = L_7 - t_{4,7} = 42.8 - 9.8 = 33$
$L_3 = L_6 - t_{3,6} = 38.8 - 9 = 29.8$
$L_1 = \text{Min } \{L_j - t_{1j}\}$
$L_2 = \text{Min } \{L_j - t_{2,j}\}$
$= \text{Min } \{L_2 - t_{1,2};\ L_3 - t_{1,3}\,;\, L_4 - t_{1,4}\}$
$= \text{Min } \{L_5 - t_{2,5}\,;\, L_6 - t_{2,6}\}$
$= \text{Min } \{16.8 - 7.8\,;\, 29.8 - 20\,;\, 33 - 33\} = 0$
$= \text{Min } \{34.8 - 18\,;\, 38.8 - 20\} = 16.8$

The *E*-value and *L*-values are shown in Fig. 13.14.

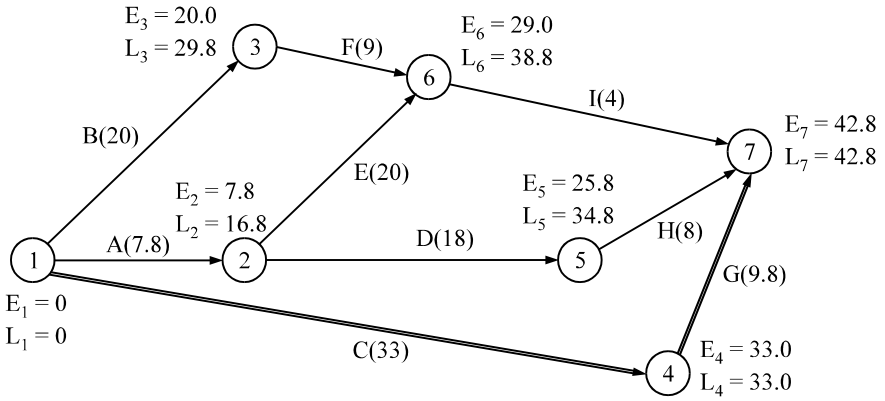


Fig. 13.14
Network Diagram

(c) The critical path is shown by thick line in Fig. 13.14 where E-values and L-values are the same. The critical path is: 1 – 4 – 7 and the expected completion time for the project is 42.8 weeks.

(d) Expected length of critical path, $T_e = t_C + t_G = 33 + 9.8 = 42.8$ weeks (Project duration).
Variance of critical path length, $\sigma^2 = \sigma_C^2 + \sigma_G^2 = 5.429 + 0.694 = 6.123$ weeks.

Since $T_s = 41.5$, $T_e = 42.8$ and $\sigma = \sqrt{6.123} = 2.474$, the probability of meeting the schedule time is given by:

$$\text{Prob}\left(Z \le \frac{T_s - T_e}{\sigma}\right) = P\left(Z \le \frac{41.5 - 42.8}{2.474}\right) = \text{Prob}\,(Z \le -0.52)$$

$$= 0.5 - 0.1952 = 0.3048 \text{ (from normal distribution table)}$$

Thus, the probability that the project can be completed in less than or equal to 41.5 weeks is 0.3048. In other words, the probability that the project will get delayed beyond 41.5 weeks is 0.6952.

Given that $$P\left(Z \le \frac{T_s - T_e}{\sigma}\right) = 0.95$$

But $Z_{0.95} = 1.64$, from normal distribution table. Thus,

$$1.64 = \frac{T_s - 42.8}{2.474} \quad \text{or} \quad T_s = 1.64 \times 2.474 + 42.8 = 46.85 \text{ weeks.}$$

Example 13.7 A small project involves 7 activities, and their time estimates are listed in the following table. Activities are identified by their beginning (i) and ending (j) node numbers.

Activity	*Estimated Duration (weeks)*		
$(i-j)$	*Optimistic*	*Most Likely*	*Pessimistic*
1 – 2	1	1	7
1 – 3	1	4	7
1 – 4	2	2	8
2 – 5	1	1	1
3 – 5	2	5	14
4 – 6	2	5	8
5 – 6	3	6	15

(a) Draw the network diagram of the activities in the project.
(b) Find the expected duration and variance for each activity. What is the expected project length?
(c) Calculate the variance and standard deviation of the project length. What is probability that the project will be completed:
  (i) at least 4 weeks earlier than expected time.
  (ii) no more than 4 weeks later than expected time.
(d) If the project due date is 19 weeks, what is the probability of not meeting the due date.

Given:

Z	:	0.50	0.67	1.00	1.33	2.00
Prob.	:	0.3085	0.2514	0.1587	0.0918	0.0228

[*Gujarat, MBA, 2001*]

Solution The network diagram of activities in the project is shown in Fig. 13.16. The earliest and latest expected time for each event is calculated by considering the expected time of each activity, as shown in Table 13.5.

Table 13.5

Activity	t_o	t_m	t_p	$t_e = \frac{1}{6}(t_o + 4t_m + t_p)$	$\sigma^2 = [\frac{1}{6}(t_p - t_o)]^2$
1 – 2	1	1	7	2	1
1 – 3	1	4	7	4	1
1 – 4	2	2	8	3	1
2 – 5	1	1	1	1	0
3 – 5	2	5	14	6	4
4 – 6	2	5	8	5	1
5 – 6	3	6	15	7	4

The E-values and L-values based on expected time (t_e) of each activity are shown in Fig. 13.15.

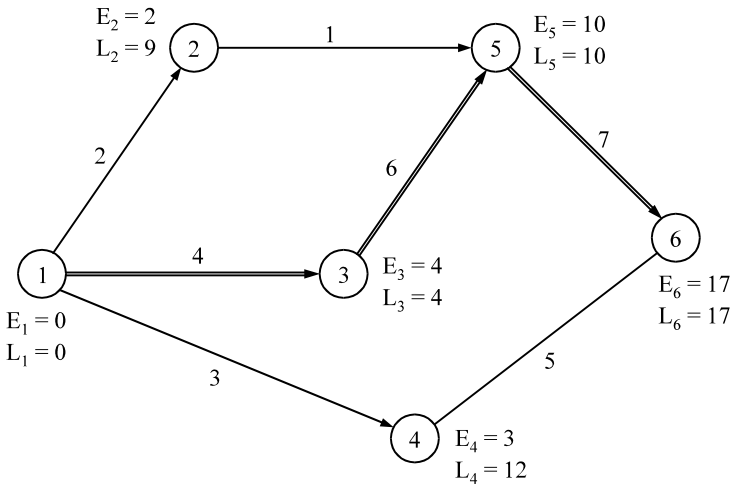


Fig. 13.15
Network Diagram

(a) Critical path is: 1 – 3 – 5 – 6.

(b) The expected duration and variance for each activity is shown in Table 13.5. The expected project length is the sum of the duration of each critical activity:

Expected project length = 1 – 3 – 5 – 6 = 4 + 6 + 7 = 17 weeks

(c) Variance of the project length is the sum of the variances of each critical activity:

Variance of project length = 1 – 3 – 5 – 6 = 1 + 4 + 4 = 9 weeks

Therefore, Standard deviation, $\sigma = \sqrt{9} = 3$

(i) Probability that the project will be completed at least 4 weeks earlier (i.e. 13 weeks) than the expected project duration of 17 weeks is given by

$$\text{Prob.} \left\{ Z \le \frac{T_s - T_e}{\sigma} = \frac{13-17}{3} \right\} = \text{Prob.} \{Z \le -1.33\} = 0.5 - 0.4082 = 0.0918$$

Thus the probability of completing the project in less than 13 days is 9.18 per cent.

(ii) Probability that the project will be completed in 4 weeks later (i.e. 21 weeks) than expected project duration of 17 weeks is given by

$$P\left\{ Z \ge \frac{21-17}{3} \right\} = P(Z \ge 1.33) = 0.5 + 0.4082 = 0.9082$$

CONCEPTUAL QUESTIONS C

1. Explain the following in the context of project management: (i) Activity variance, and (ii) Project variance.
2. Explain the following terms in PERT: (i) Three time estimates, (ii) Expected time, and (iii) Activity variance.
3. What are the basic assumptions underlying the 'expected time' estimate?
4. 'PERT takes care of uncertain durations.' How far is this statement correct? Explain with reasons.
5. How does PERT provide for uncertainty in activity time estimates? What is the rationale for using beta probability distribution? [*Delhi Univ., MBA, 2005*]

SELF PRACTICE PROBLEMS C

1. A project has the following activities and other characteristics:

Activity	*Preceding Activity*	*Time Estimates (weeks)*		
		Optimistic	*Most Likely*	*Pessimistic*
A	–	4	7	16
B	–	1	5	15
C	*A*	6	12	30
D	*A*	2	5	8
E	*C*	5	11	17
F	*D*	3	6	15
G	*B*	3	9	27
H	*E, F*	1	4	7
I	*G*	4	19	28

(a) Draw the network diagram for the project.
(b) Identify the critical path.
(c) Prepare the activity schedule for the project.
(d) Determine the mean project completion time.
(e) Find the probability that the project is completed in 36 weeks. [*Delhi Univ., MCom, 2001*]

2. A small project consists of seven activities, the details of which are given below:

Activity	Duration (days)			Immediate Predecessor
	Most Likely	Optimistic	Pessimistic	
A	3	1	7	–
B	6	2	14	A
C	3	3	3	A
D	10	4	22	B, C
E	7	3	15	B
F	5	2	14	D, E
G	4	4	4	D

(a) Draw the network, number the nodes, find the critical path, the expected project completion time and the next most critical path.
(b) What project duration will have 95 per cent confidence of completion?

3. The owner of a chain of fast food restaurants is considering a new computer system for accounting and inventory control. A computer company sent the following information about the computer system installation:

Activity	Activity Description	Immediate Predecessor	Times (days)		
			Optimistic	Most Likely	Pessimistic
A	Select the computer model	–	4	6	8
B	Design input/output system	A	5	7	15
C	Design monitoring systems	A	4	8	12
D	Assemble computer hardware	B	15	20	25
E	Develop the main programmes	B	10	18	26
F	Develop input/output routines	C	8	9	16
G	Create database	E	4	8	12
H	Install the system	D, F	1	2	3
I	Test and implement	G, H	6	7	8

(a) Construct the network diagram for the project.
(b) Determine the critical path and compute the expected completion time.
(c) Determine the probability of completing the project in 55 days.

4. A company manufacturing plant and equipment for chemical processing is in the process of quoting a tender called by a public sector undertaking. The delivery data, once promised, is crucial and a penalty clause is applicable. The project manager has listed down the activities in the project as under:

Activity	Immediate Predecessor	Activity Time (weeks)		
		Optimistic	Most Likely	Pessimistic
A	–	1	3	5
B	–	2	4	6
C	A	3	5	7
D	A	5	6	7
E	C	5	7	9
F	D	6	8	10
G	B	7	9	11
H	E, F, G	2	3	4

(a) Find out the delivery week from the date of commencement of the project, and
(b) Find the total float and free float for each of the non-critical activities.

5. Consider a project having the following activities and their time estimates:

Activity	Immediate Predecessor	Activity Time (weeks)		
		Most Optimistic	Most Likely	Most Pessimistic
A	–	3	4	5
B	–	4	8	10
C	B	5	6	8
D	A, C	9	15	10
E	B	4	6	8
F	D, E	3	4	5
G	D, E	5	6	8
H	D, E	1	3	4
I	G	2	4	5
J	F, I	7	8	10
K	G	4	5	6
L	H	8	9	13
M	J, K, L	6	7	8

(a) Draw the network diagram for the project.
(b) Compute the expected project completion time.
(c) What should be the due date to have 0.90 probability of completion?
(d) Find the total float and free float for all non-critical activities.
[Agra, Univ., MCA, 2000]

6. A sociologist plans a questionnaire survey, consisting of the following tasks:

Activity	Description	Immediate Predecessor	Duration (days)		
			Likely	Minimum	Maximum
A	Design of questionnaire	–	5	4	6
B	Sampling design	–	12	8	16
C	Testing of questionnaire and refinements	A	5	4	12
D	Recruiting for interviewers	B	3	1	5
E	Training of interviewers	D, A	2	2	2
F	Allocation of areas to interviewers	B	5	4	6
G	Conducting interviews	C, E, F	14	10	18
H	Evaluation of results	G	20	18	34

(a) For this PERT network find the expected task durations and the variances of task durations.
(b) Draw a network for this project and find the critical path. What is the expected length of the critical path? What is the variance of the length of the critical path?
(c) What is the probability that the length of the critical path does not exceed 60 days? [*Delhi Univ., MBA, 2002*]

7. A management student identifies the following list of activities and sequencing requirements along with the time estimates for various activities related to the completion of his project work.

Activity	Description	Immediate Predecessor	Activity Time (days)		
			Optimistic	Likely	Pessimistic
A	Search of list	–	3	6	9
B	Dedicating the project	–	2	4	12
C	Preliminary work	B	1	1.5	5
D	Formal proposal	C	1	2	3
E	Project committee's approval	A, D	1.5	2	4.5
F	Progress report	E	0.5	1	1.5
G	Format research	A, D	4.5	5	11.5
H	Data collection	E	2	5	8
I	Analysis	G, H	4	5.5	10
J	Conclusion	I	1.5	2.5	4.5

K	Draft	*I, F*	2	3.5	8
L	Final draft	*J, K*	2.5	3	1.5
M	Presentation	*L*	0.5	1	1.5

With the help of a network diagram, determine the minimum time required to complete the project. From the network, identify the activities that can be delayed without affecting the project duration and the extent of delay that is possible for such activities. [*Delhi Univ., MBA, 2004*]

8. A publisher is preparing to produce the second edition of a textbook. The activities required and their estimated time are as follows:

Activity	*Description*	*Immediate Predecessor*	*Activity Time (days)*		
			Optimistic	*Likely*	*Pessimistic*
A	Assess market	–	1	3	2
B	Get reviews from users	*A*	1	2	1.5
C	Revamp old material and add new material	*B*	3	9	5
D	Obtain review and prepare final draft	*C*	4	12	6
E	Revise and expand problems	*B*	2	7	4
F	Copy edit final draft	*D*	1	2.5	1.5
G	Copy edit problems	*E*	0.5	1.5	1
H	Set type, proof and print book	*F, G*	5	9	6
I	Prepare instructor's manual	*E*	2	4	3
J	Produce instructor's manual	*I*	1	2	1.5
K	Complete book and instructor's manual	*H, J*	0	0	0

(a) Draw a PERT network diagram and determine the critical path.
(b) What is the probability that the project will be completed within (i) 21 months? (ii) 27 months?

9. A nationalized bank wishes to plan and schedule the development and installation of a new computerized cheque processing system. The changeover in cheque-processing procedures requires employment of additional personnel to operate the new system, development of new systems (computer software), and modification of existing cheque sorting equipment. The activities required to complete the project, along with three time estimates and the precedence relationship among the activities, have been determined by the bank management and are given in the following table:

Activity	*Description*	*Predecessor*	*Time Estimates (days)*		
			Optimistic	*Most Likely*	*Pessimistic*
A	Position recruiting	–	5	8	17
B	System development	–	3	12	15
C	System training	*A*	4	7	10
D	Equipment training	*A*	5	8	23
E	Manual system test	*B, C*	1	1	1
F	Preliminary system changeover	*B, C*	1	4	13
G	Computer-personnel interface	*D, E*	3	6	9
H	Equipment modification	*D, E*	1	2.5	7
I	Equipment testing	*H*	1	1	1
J	System debugging and installation	*F, G*	2	2	2
K	Equipment changeover	*G, I*	5	8	11

(a) Draw the network diagram for this project and find the critical path and its length.
(b) Calculate the total and free floats for non-critical activities.
(c) What is the probability that the length of the critical path does not exceed 40 days? [*Delhi Univ., MBA, 2003*]

10. A promoter is organizing a sports meeting. The relationship among the activities and time estimates in days are shown below in the table:

Activity	*Description*	*Predecessor*	*Activity Time (days)*		
			Optimistic	*Likely*	*Pessimistic*
A	Prepare draft programme	–	3	7	11
B	Send to sports organizations and wait for comments	*A*	14	21	28
C	Obtain promoters	*A*	11	14	17
D	Prepare and sign documents for stadium hire	*A, C*	2	2	2
E	Redraft programme and request entries	*B*	2	3.5	8
F	Enlist officials	*D, E*	10	14	21
G	Arrange accommodation for touring teams	*E*	3	4	5
H	Prepare detailed programme	*E, F*	4	4.5	8
I	Make last-minute arrangements	*G, H*	1	2	4

(a) Draw the network diagram for the project and compute the expected completion time of the project.
(b) What should be the due date to have 0.90 probability of project completion?
(c) Find the total float and free float for all non-critical activities.
(d) What is the probability that the length of the critical path does not exceed 56 days? [*Delhi Univ., MBA, 2005*]

11. A management student identifies the following list of activities and sequencing requirements along with the time estimates for various activities related to the completion of his project:

Activity	*Description*	*Immediate Predecessor*	*Activity Time (days)*		
			Optimistic	*Likely*	*Pessimistic*
A	Search of Literature	–	3	6	9
B	Deciding the project	–	2	4	12
C	Preliminary work	*B*	1	1.5	5
D	Formal proposal	*C*	1	2	3
E	Project committee's approval	*A, D*	1.5	2	4.5
F	Progress report	*E*	0.5	1	1.5
G	Formal research	*A, D*	4.5	5	11.5
H	Data collection	*E*	2	5	8
I	Analysis	*G, H*	4	5.5	10
J	Conclusion	*I*	1.5	2.5	4.5
K	Draft	*I, F*	2	3.4	8
L	Final Draft	*J, K*	2.5	3	1.4
M	Presentation	*L*	0.5	1	1.5

With the help of an arrow diagram, determine the minimum time required to complete the project. From the network, identify the activities that can be delayed without affecting the project duration and the extent of delay that is possible for such activities. [*Delhi Univ., MBA, 2001*]

12. A computer software company has broken down the process of integrating a computer system into its operation into several steps. Some of the steps cannot begin until the others are completed. These relationships are shown in the accompanying table. In addition, estimates of the most likely, optimistic, and pessimistic times required for each are listed below:

Activity	*Immediate*	*Expected Time (weeks)*		
		Optimistic	*Most Likely*	*Pessimistic*
A	–	2	3	4
B	–	3	4	11
C	*A*	2	5	8
D	*A*	1.5	3.5	8.5
E	*B*	5	7	9
F	*B*	2	5.5	6
G	*C*	1.5	2.5	6.5
H	*C, D*	3	4	11
I	*G*	4	6	8
J	*H, E*	3	4.5	9
K	*F*	5	6	7
L	*I, J, K*	1	3	11

(a) Draw a PERT chart for this project, and calculate the critical path.
(b) By how much time can activity F be delayed without delaying the project as a whole?
(c) If labour costs Rs 1,500 per week find the probability that the labour costs for this project will exceed Rs 36,000.
(d) Company wishes to budget an amount for labour costs that will be sufficient with 95 per cent probability. How much should they budget?

13. A multinational FMCG company wishes to launch a new Fruit Yogurt in the coming season. A brief description of the activities associated with this project, their expected durations (in weeks) and their immediate predecessor(s) are given in the following table:

Activity	*Description*	*Predecessor*	*Expected Time (weeks)*		
			Optimistic	*Likely*	*Pessimistic*
A	Management approval	—	2	2.5	4
B	Product concept test	*A*	3	4.7	5
C	Technical feasibility	*A*	2	2	3
D	Recipe finalization	*C, B*	1	1	2
E	Shelf life trials	*D*	8	12	15
F	Brand positioning study	*B*	4	5	7
G	Packaging key lines	*F*	1	1	2.5
H	Agency advertisement development	*F, E*	4	8	10
I	Agency: layouts artworks	*G*	2	3	5
J	Advertisement test research	*H*	3	4	5
K	Cost finalization	*D*	1	1	2
L	Pricing decision	*J, F*	1	1.5	3
M	Marketing mix finalization	*L*	2	2.5	3
N	POS development	*M*	2	4	5
O	Launch plans	*M*	1	1	1.5
P	Branch communication	*O*	0.5	0.5	1
Q	Supplier's delivery of packaging	*H*	4	5	8
R	Production trail	*D, Q*	1.5	2	3
S	Management final approval	*R*	0.5	1	1.5
T	Final production	*S*	1	2.5	3
U	Stock movement	*T*	0.5	1.5	2
V	Position movement	*T*	0.5	1	1.5
W	Launch	*U, V*	1	2	4

The management of the company desires to know the realistic completion time for this project and detailed analysis of float times (if any). [*Delhi Univ., MBA, 2000, 2004*]

14. Following is the list of various activities involved in the launch of a new Credit Card service by a company, their immediate predecessors and their expected durations (in days):

Activity	*Description*	*Immediate Predecessor*	*Activity Time (days)*		
			Optimistic	*Likely*	*Pessimistic*
A	Conduct market research to determine size, potential and competition in the market	–	10	12	14
B	Define marketing strategy in terms of positioning and product features	*A*	14	15	17
C	Estimate expected volumes	*B*	2	3	4
D	Estimate additional manpower required	*C*	4	6	8
E	Identify modifications required to the present computer system based on expected volumes	*C*	10	12	14
F	Implement computer system modifications	*E*	20	25	27
G	Update physical facilities	*C*	10	17	20
H	Prepare instruction manuals for modified system	*F*	5	6	7
I	Hire additional sales force and operations staff	*D*	7	12	14
J	Train sales force and operations staff	*H, I*	14	17	20
K	Advertise for suppliers	*C*	1	2	3
L	Check supplier samples	*K*	10	15	20
M	Identify key suppliers and define quality standards	*L*	3	5	7
N	Roll-out basic product to a few employees as a method of pre-testing	*M, J*	13	15	17
O	Generate and assess feedback	*N*	20	21	22
P	Modify and make changes wherever necessary	*O*	7	9	14
Q	Update procedures manual	*P*	2	3	4
R	Retain commercial staff	*Q*	2	2	2
S	Design communications to prospective card-holders	*P*	7	10	13
T	Design advertising schedule	*S*	5	7	9
U	Roll out final product-Launch	*T, R, G*	4	8	12

(a) Draw the network diagram for the project.
(b) Find the expected project completion time of the project.
(c) Determine the probability of completing the project in 165 days. [*Delhi Univ., MBA, 2000*]

HINTS AND ANSWERS

1. (i) Critical path: A – C – E – H. Expected project duration is 37 weeks with its variance 25 weeks.

   (ii) $Z = \frac{T_s - T_e}{\sigma} = \frac{36-37}{5} = -0.2$; $P(Z \le -0.2) = 0.5 - 0.0793 = 0.4207$.

   (iv) $Z(= 2.33) = \frac{T_s - 37}{5}$ or $T_s = 37 + 2.33 \times 5 = 48.65$ weeks.

2. (i) Critical path A – B – D – F. Expected project duration is 27 days. Next critical path: A – B – D – G with duration of 25 days.

   (ii) Project duration which has 95 per cent probability of completion is

   $$Z\ (= 1.64) = \frac{T_s - 27}{\sqrt{18}}$$

   or $T_s = 1.64 \times \sqrt{18} + 27 = 34$ days (approx)

3. (i) Critical path: A – B – E – G – I with project duration of 47 days and its variance is 110/9.

   (ii) Probability of completing the project in 55 days is

   $$P\left[Z \le \frac{T_s - T_e}{\sigma} = \frac{55-47}{3.49}\right] = P(Z \le 2.29) = 0.9634$$

13.7 PROJECT TIME-COST TRADE-OFF

While discussing the scheduling of project activities in previous section, the cost of resources consumed by activities were not taken into consideration. The project completion time can be reduced by reducing (crashing) the normal completion time of critical activities. The crashing of normal completion time of critical activities will increase the total cost of the project. But, the decision-maker will always look for trade-off between the total cost of project and the total time required to complete it.

Crashing the project means crashing a number of activities to reduce the duration of the project, below its normal time.

13.7.1 Project Crashing

Crashing of completion time of critical activities to reduce the project completion time would certainly require extra resources (cost). However, as shown in Fig. 13.16, beyond point *A* project cost increases when time is reduced. Similarly, beyond point *B*, the time increases while the project cost decreases. Obviously, an activity time may not be reduced beyond a certain limit and this limit is referred as *crash point.* Also, extending an activity time beyond normal point (cost efficient) may increase cost of executing that activity.

For simplicity, the relationship between normal time and cost as well as crash time and cost for an activity is assumed to be linear. Thus, the crash cost per unit of time can be estimated by computing the relative change in cost (cost slope) per unit change in time.

As shown in Fig. 13.16, a decision-maker must be interested in the central region of the curve between points *A* and *B*. This will help in developing a trade-off between time and direct cost of completing an activity.

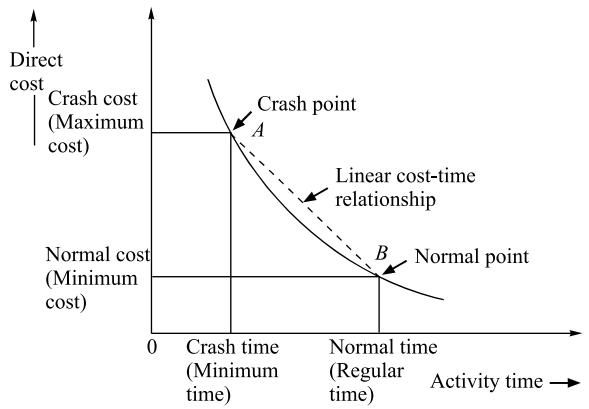


Fig. 13.16 Time cost Trade-off

13.7.2 Time-Cost Trade-Off Procedure

The starting point for crashing is when all critical activities are completed with their normal time, and crashing is stopped when all critical activities are crashed.

The method of establishing time-cost trade-off for the completion of a project can be summarized as follows:

Step 1: Determine the normal project completion time and associated critical path.

Step 2: Identify critical activities and compute the cost slope for each of these by using the relationship

$$\text{Cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

The values of cost slope for critical activities indicate the direct extra cost required to execute an activity per unit of time.

Crashing an activity means taking special costly measures to reduce the duration of an activity below its normal time.

Step 3: For reducing the total project completion time, identify and crash an activity time on the critical path with lowest cost slope value to the point where

(i) another path in the network becomes critical, or
(ii) the activity has been crash to its lowest possible time.

Step 4: If the critical path under crashing is still critical, return to Step 3. However, if due to crashing of an activity time in Step 3, other path(s) in the network also become critical, then identify and crash the activity(s) on the critical path(s) with the minimum joint cost slope.

Step 5: Terminate the procedure when each critical activity has been crashed to its lowest possible time. Determine total project cost (indirect cost plus direct cost) corresponding to different project durations.

Example 13.8 The data on normal time, and cost and crash time and cost associated with a project are shown in the following table.

Activity	*Normal*		*Crash*	
	Time(*weeks*)	*Cost* (*Rs*)	*Time* (*weeks*)	*Cost* (*Rs*)
1 – 2	3	300	2	400
2 – 3	3	30	3	30
2 – 4	7	420	5	580
2 – 5	9	720	7	810
3 – 5	5	250	4	300
4 – 5	0	0	0	0
5 – 6	6	320	4	410
6 – 7	4	400	3	470
6 – 8	13	780	10	900
7 – 8	10	1,000	9	1,200
		4,220		

Indirect cost is Rs 50 per week.

(a) Draw the network diagram for the project and identify the critical path.
(b) What are the normal project duration and associated cost?
(c) Find out the total float associated with non-critical activities.
(d) Crash the relevant activities and determine the optimal project completion time and cost.

Solution (a) The network for normal activity times is shown in Fig. 13.17. The critical path is: 1 – 2 – 5 – 6 – 7 – 8 with a project completion time of 32 weeks.

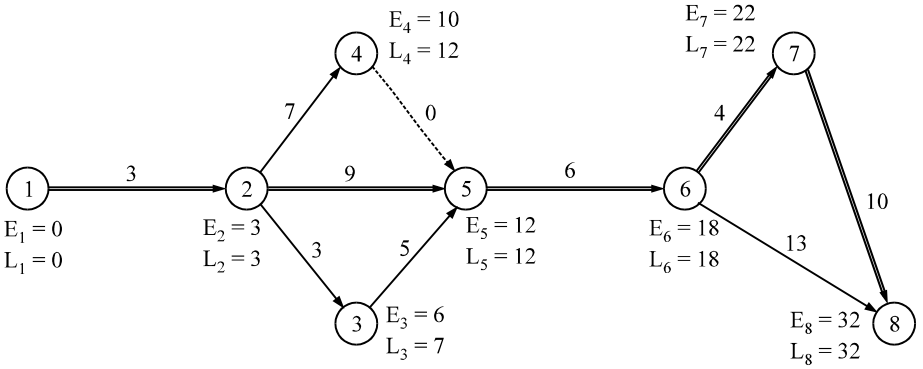


Fig. 13.17 Network Diagram

(b) The normal total project cost associated with normal project duration of 32 weeks is as follows:

Total cost = Direct normal cost + Indirect cost for 32 weeks

= 4,220 + 50 × 32 = Rs 5,820

(c) Calculations for total float associated with non-critical activities are shown in Table 13.6.

Activity	*Total Float* $(L_j - E_i) - t_{ij}$
2 – 3	(7 – 3) – 3 = 1
2 – 4	(12 – 3) – 7 = 2
3 – 5	(12 – 6) – 5 = 1
4 – 5	(12 – 10) – 0 = 2
6 – 8	(32 – 18) – 13 = 1

Table 13.6
Total Float

(d) For critical activities, crash cost-slope is given in Table 13.7.

Critical Activity	*Crash Cost per Week (Rs)*
1 – 2	$\frac{400-300}{3-2} = 100$
2 – 5	$\frac{810-720}{9-7} = 45$
5 – 6	$\frac{410-320}{6-4} = 45$
6 – 7	$\frac{470-400}{4-3} = 70$
7 – 8	$\frac{1200-1000}{10-9} = 200$

Table 13.7
Crash Cost Slope

The minimum value of crash cost per week is for activity 2 – 5 and 5 – 6. Hence, crashing activity 2 – 5 by 2 days from 9 weeks to 7 weeks. But the time should only be reduced by 1 week otherwise another path 1 – 2 – 3 – 5 – 6 – 7 – 8 become a parallel path. Network, as shown in Fig. 13.18, is developed when it is observed that new project time is 31 weeks and the critical paths are 1 – 2 – 5 – 6 – 7 – 8 and 1 – 2 – 3 – 5 – 6 – 7 – 8.

With crashing of activity 2 – 5, the crashed total project cost becomes:

Crashed total cost = Total direct normal cost + Increased direct cost due to crashing of activity (2 – 5) + Indirect cost for 31 weeks

= 4,220 + 1 × 45 + 50 × 31 = 4,265 + 1,550 = Rs 5,815

For revised network shown in Fig. 13.18, new possibilities for crashing critical activities are listed in Table 13.8.

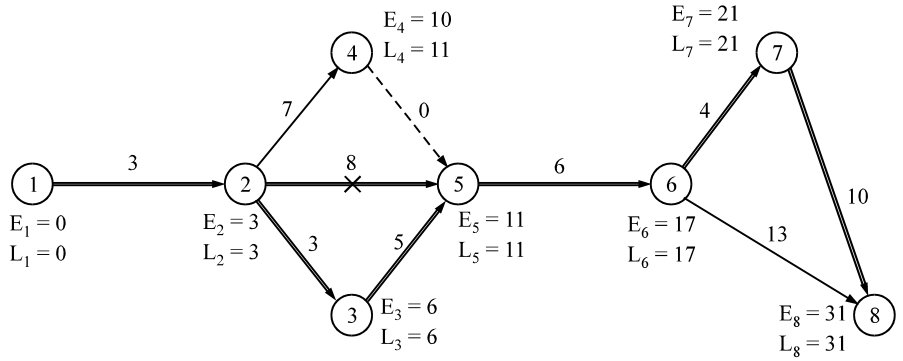


Fig. 13.18
Network Diagram

Critical Activity	*Crashed Cost per Week (Rs)*
1 – 2	$\frac{400-300}{3-2}=100$
2 – 5	× (Crashed)
2 – 3	0 (Crashing is not required)
3 – 5	$\frac{300-250}{5-4}=50$
5 – 6	$\frac{410-320}{6-4}=45$
6 – 7	$\frac{470-400}{4-3}=70$
7 – 8	$\frac{1200-1000}{10-9}=200$

Table 13.8
Crash Cost Slope

Since crashed cost slope for activity 5 – 6 is minimum, its time may be crashed by 2 weeks from 6 weeks to 4 weeks. The updated network diagram is shown in Fig. 13.19.

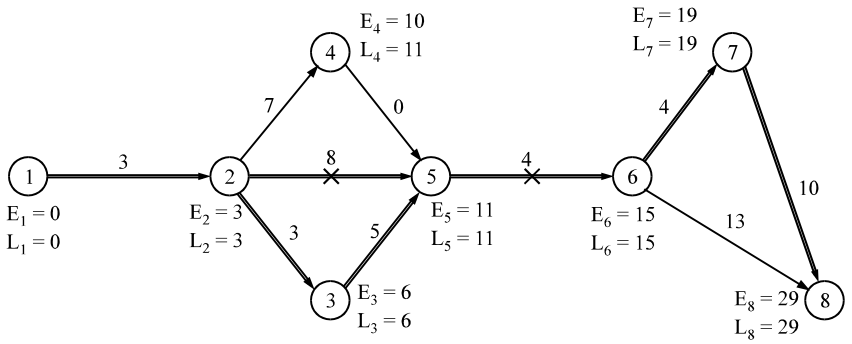


Fig. 13.19
Network Diagram

It may be noted in Fig. 13.19, that the critical paths shown in Fig. 13.18 remain unchanged because activity 5 – 6 is common in both. With crashing of activity 5 – 6 by 2 weeks, the crashed total cost becomes:

Crashed total cost = Total direct normal cost + Increased direct cost due to crashing of 5 – 6 + Indirect cost for 29 weeks

$= 4{,}220 + (1 \times 45 + 2 \times 45) + 50 \times 29 =$ Rs 5,805

For revised network given in Fig 13.19, new possibilities for crashing in the critical paths are listed in Table 13.9.

Critical Activity	*Crashed Cost per Week (Rs)*
1 – 2	$\frac{400-300}{3-2}=100$
2 – 3	0 (Crashing is not required)
2 – 5	× (crashed)
5 – 6	× (crashed)
6 – 7	$\frac{470-400}{4-3}=70$
7 – 8	$\frac{1200-1000}{10-9}=200$

Table 13.9
Crash Cost Slope

The further crashing of 6 – 7 activity time from 4 weeks to 3 weeks will result in increased direct cost than the gain due to reduction in project time. Hence, terminate crashing. The optimal project duration is 29 weeks with associated cost of Rs 5,805 as shown in Table 13.10.

Project Duration (weeks)	Crashing Activity and Weeks	Direct Cost (Rs)			Indirect Cost (Rs)	Total Cost (Rs)
		Normal	Crashing	Total		
32	–	4,220	–	4,220	32 × 50 = 1,600	5,820
31	2 – 5(1)	4,220	1 × 45 = 45	4,265	31 × 50 = 1,550	5,815
29	5 – 6(2)	4,220	45 + 2 × 45 = 135	4,355	29 × 50 = 1,450	5,805
28	6 – 7(1)	4,220	135 + 1 × 70 = 205	4,425	28 × 50 = 1,400	5,825

Table 13.10 Crashing Schedule of Project

Example 13.9 The data on normal time and cost along with crashed time and cost associated with a project are shown in the following table:

Acltivity	Immediate Predecessor	Normal		Crash	
		Time (weeks)	Cost (Rs '000)	Time (weeks)	Cost (Rs '000)
A	–	10	20	7	30
B	–	8	15	6	20
C	B	5	10	4	14
D	B	6	11	4	15
E	B	8	9	5	15
F	E	5	5	4	8
G	A, D, C	12	3	8	4
			71,000		

The indirect cost is Rs. 400 per day. Find the optimum duration and the associated minimum project cost.

Solution The network diagram with the normal time of the project activities is shown in Fig. 13.20. The critical path in the network diagram is shown by thick lines. The critical path is: 1 – 2 – 3 – 4 – 6 with normal project duration of 26 weeks.

Project cost associated with project duration of 26 weeks is Rs 1,43,800:

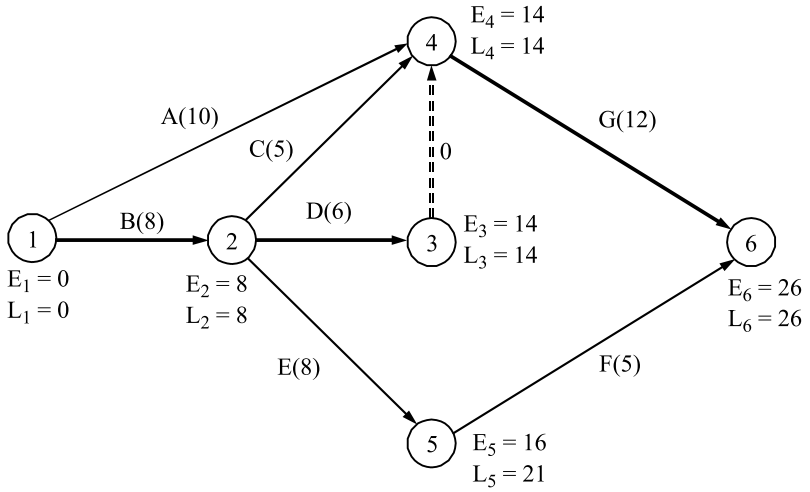


Fig. 13.20 Network Diagram

Total cost = Direct normal cost + Indirect cost for 26 weeks
= 71,000 + 26 × 7 × 400 = Rs 1,43,800

To begin crash analysis, the crash cost slope values for critical activities is shown in Table 13.11.

Critical Activity	Crash Cost per Week (Rs)
B (1 – 2)	$\frac{20-15}{8-6} = 2.5$
D (2 – 3)	$\frac{15-11}{6-4} = 2$
G (4 – 6)	$\frac{4-3}{12-8} = 0.25$

Table 13.11 Crash Cost Slope

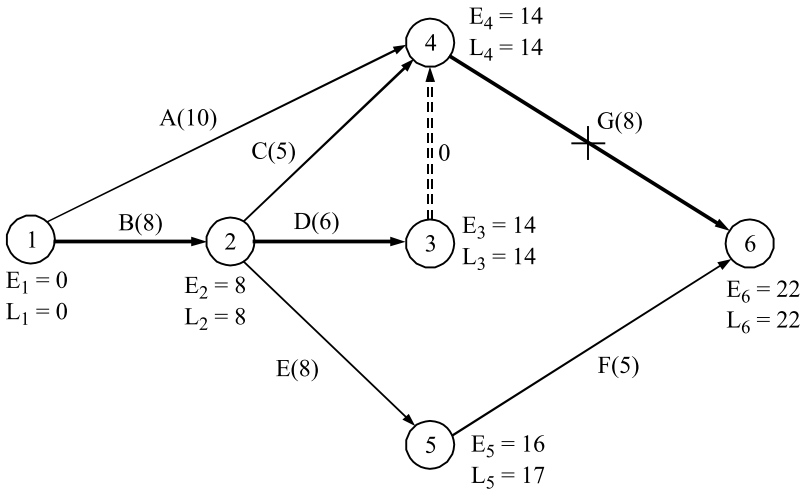


Fig. 13.21
Network Diagram

The critical activity *G* with cost slope of Rs (0.25 × 1,000) = Rs 250 per week is lowest and can be crashed by 4 weeks (being last activity of critical path). The revised network is shown in Fig. 13.21. The new critical path is still 1 – 2 – 3 – 4 – 6 with total project duration of 22 weeks.
With crashing of activity *G* (4 – 6), the total project cost becomes:

Total crashed cost = Direct normal cost + Increased cost due to crashing of *G* + Indirect cost for 22 weeks

= 71,000 + 0.25 × 1000 × 4 + 22 × 7 × 400

= 71,000 + 1000 + 61,600 = Rs. 1,33,600

In Fig. 13.21, the path 1 – 2 – 3 – 4 – 6 is still the critical path. As shown in Table 13.11, the next least expensive activity on the critical path is activity *D* (2 – 3). It can be crashed by one week. The crashed network is shown in Fig. 13.22 with total project duration of 21 weeks.

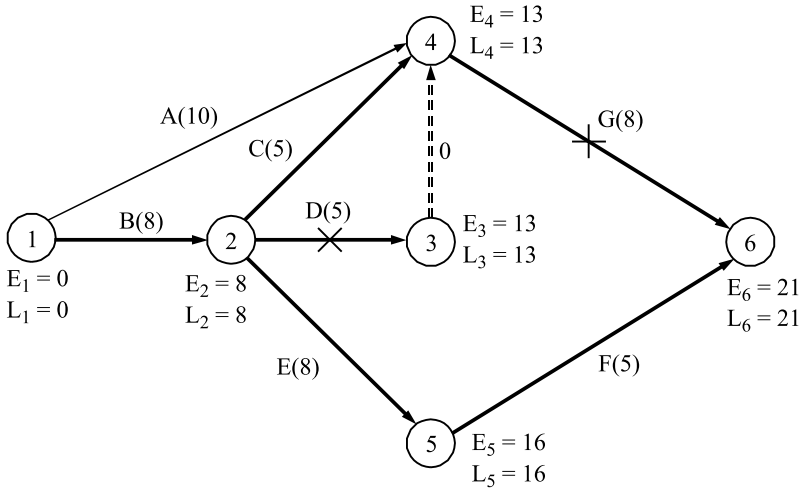


Fig. 13.22
Network Diagram

With the crashing of activity *D*, the total project duration is reduced to 21 weeks with the critical paths each with project duration of 21 weeks are shown in Fig. 13.22.

(i) 1 – 2 – 5 – 6
(ii) 1 – 2 – 3 – 4 – 6
(iii) 1 – 2 – 4 – 6

The new total project cost involved with network shown in Fig. 13.22 becomes:

Total crashed cost = Direct normal cost + Increased direct cost due to crashing of *D* + Indirect cost for 21 weeks

= 71,000 + (1000 + 2 × 1000 × 1) + 21 × 7 × 400

= 71,000 + 1000 + 2000 + 58,800 = Rs 1,32,800

Since activity *B* is common in all three critical paths, as shown in Fig. 13.22, therefore it can be crashed to a maximum of 2 weeks. The new total project cost with project duration of 19 weeks becomes:

Total crashed cost = Direct normal cost + Increased direct cost due to crashing of *B*, *D* and *E* + Indirect cost for 19 weeks

$= 71{,}000 + (1{,}000 + 2{,}000 + 2.5 \times 1{,}000 \times 2) + 19 \times 7 \times 400$

$= 73{,}000 + 5{,}000 + 53{,}200 = \text{Rs } 1{,}31{,}200$

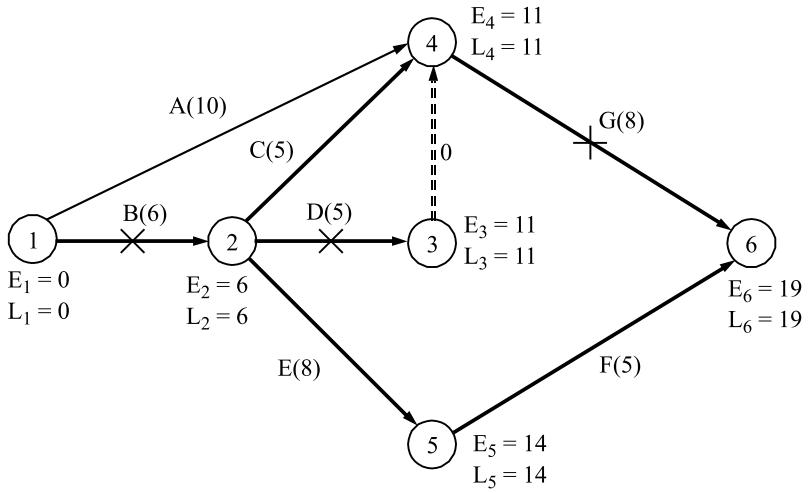


Fig. 13.23
Network Diagram

With the crashing of activity *B* by 2 weeks, the project duration is reduced to 19 weeks with the following critical paths each with project duration of 19 weeks as shown in Fig. 13.23.

(i) 1 – 2 – 5 – 6
(ii) 1 – 2 – 3 – 4 – 6
(iii) 1 – 2 – 4 – 6

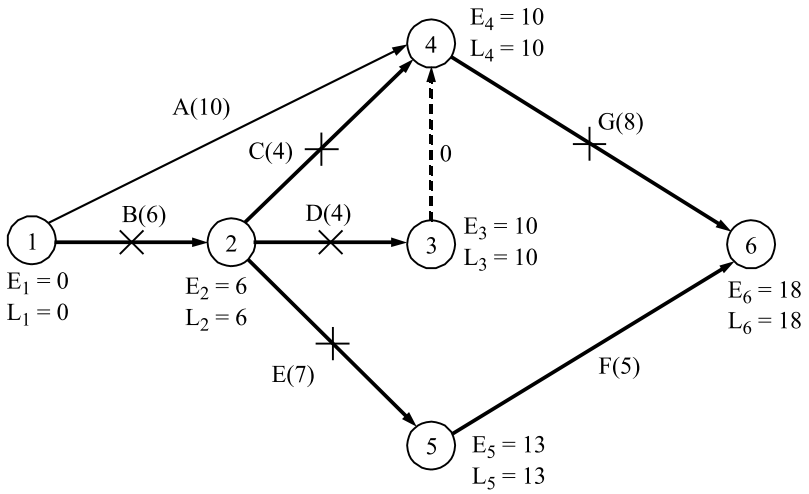


Fig. 13.24
Network Diagram

For revised network shown in Fig. 13.24, the crashing possibilities on the critical path are listed in Table 13.12.

Crashed Activity	*Creashed Cost per Week (Rs)*
2 – 4	$\frac{14-10}{5-4}=4$
2 – 3	$\frac{15-11}{6-4}=2$
2 – 5	$\frac{15-9}{8-5}=2$

Table 13.12
Crash Cost slope

Since critical activities B and G cannot be crashed further, therefore crashing activity C in path 1 – 2 – 4 – 6, activity D in path 1 – 2 – 3 – 4 – 6 and activity E in path 1 – 2 – 5 – 6 each by 1 week in order to reduce the project duration to 18 weeks which is equivalent to the project time of non-critical path 1 – 4 – 6. The reduced project duration along with critical paths is shown in Fig. 13.24.

The total project cost with project duration of 18 weeks becomes:

Total crashed cost = Direct normal cost + Increased direct cost due to crashing of activities C, D and E + Indirect cost for 18 weeks

$= 71{,}000 + (1{,}000 + 2{,}000 + 5{,}000) + (4 \times 1{,}000 \times 1 + 2 \times 1{,}000 \times 1 + 2 \times 1{,}000 \times 1) + 7 \times 400 \times 18 = 1{,}37{,}400$

Since total project cost for 18 weeks is more than the cost for 19 weeks, therefore, further crashing is not desirable. Hence, the optimum pair of time and cost associated with the project is 19 weeks and Rs 1,32,200, as shown in Table 13.13.

Table 13.13 Crashing Schedule of the Project

Project Duration (weeks)	*Crashing Activity and Weeks*	*Direct Cost (Rs)*			*Indirect Cost (Rs)*	*Total Cost (Rs)*
		Normal	*Crashing*	*Total*		
26	–	71,000	–	71,000	26 × 2,800 = 72,800	1,43,800
22	4 – 6(4)	71,000	4 × 250 = 1,000	72,000	22 × 2,800 = 61,600	1,33,600
21	2 – 3(1)	71,000	1,000 + 2,000 = 3,000	74,000	21 × 2,800 = 58,800	1,32,800
19	1 – 2(2)	71,000	3,000 + 5,000 = 8,000	79,000	19 × 2,800 = 53,200	**1,32,200**
18	2 – 4(1), 2 – 3(1), 2 – 5(1)	71,000	8,000 + 8,000 = 16,000	87,000	18 × 2,800 = 50,400	1,37,400

Example 13.10 The following table gives the activities in a construction project and also gives other relevant information:

Activity	*Immediate Predecessor*	*Time (months)*		*Direct Cost (Rs '000)*	
		Normal	*Crash*	*Normal*	*Crash*
A	–	4	3	60	90
B	–	6	4	150	250
C	–	2	1	38	60
D	*A*	5	3	150	250
E	*C*	2	2	100	100
F	*A*	7	5	115	175
G	*D, B, E*	4	2	100	240
				713	

Indirect costs vary as follows:

Months	:	15	14	13	12	11	10	9	8	7	6
Cost (Rs)	:	600	500	400	250	175	100	75	50	35	25

(a) Draw an arrow diagram for the project.

(b) Determine the project duration that will result in minimum total project cost. [*Osmania Unit., MBA, 2002*]

Solution The network for normal activity times indicates a project duration of 13 months with critical path: A – D – G (1 – 2 – 4 – 5) as shown in Fig. 13.25. The crash cost slope for various critical activities of the project is given in Table 13.14.

Table 13.14 Crash Cost Slope

Critical Activity	*Crash cost/month (Rs)*
$A(1-2)$	$\frac{90-60}{4-3} = 30$
$D(2-4)$	$\frac{250-150}{5-3} = 50$
$G(4-5)$	$\frac{240-100}{4-2} = 70$

Among the critical activities *A*, *D* and *G*, the least expensive activity is *A*. This means that the duration of this activity should be crashed by maximum of one month. Thus, project duration reduces to 12 months and new total project cost is as follows:

Total crashed cost = Total direct cost + Indirect cost due to crashing of activity *A* by one week + Indirect cost for 12 months

= 713 + 1 × 30 + 250 = Rs 993

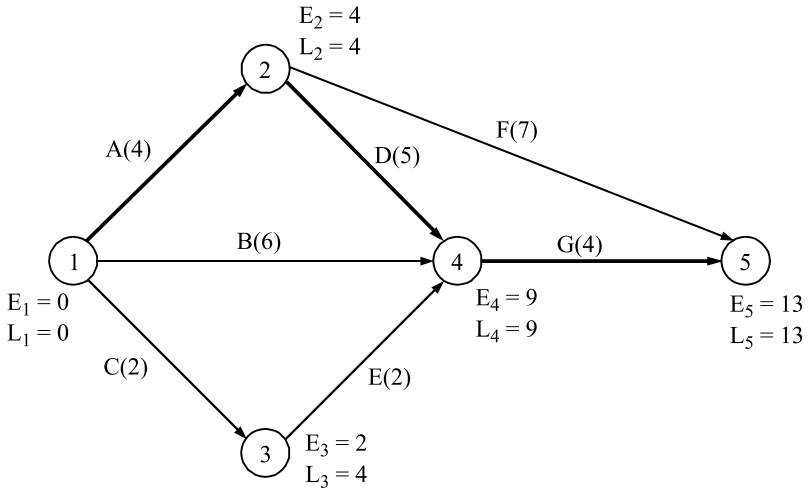


Fig. 13.25
Network Diagram

Since existing critical path remains same even after crashing critical activity A by one month, therefore next lowest cost slope activity *D* is crashed by 2 months. Thus, project duration reduces to 10 months as shown in Fig. 13.26.

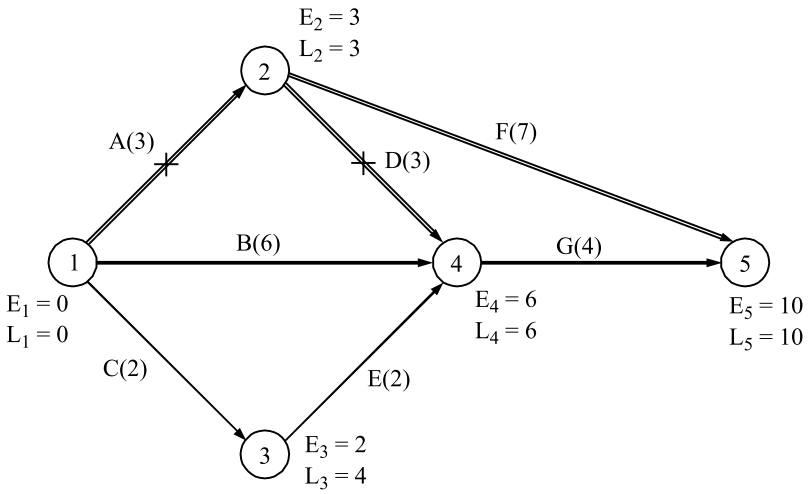


Fig. 13.26
Network Diagram

In Fig. 13.26, there are three critical paths: 1 – 2 – 5; 1 – 2 – 4 – 5 and 1 – 4 – 5, each having a duration of 10 months. The new total project cost becomes:

Total crashed cost = Total direct cost + Indirect cost due to crashing of activity *D* by 2 months + Indirect cost for 10 months

= 713 + (30 + 2 × 50) + 100 = Rs 943

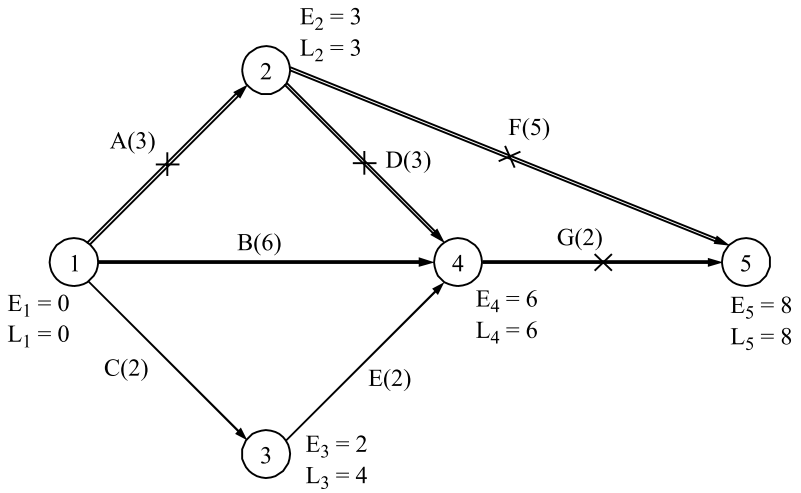


Fig. 13.27 Network Diagram

Since the critical activities *A* and *D* cannot be further crashed, therefore activities *F* and *G* may be crashed by 2 months each. The revised project network is shown in Fig. 13.27 with total project duration of 8 months. The new total project cost then becomes:

Total crashed cost = Total direct cost + Increased direct cost due to crashing of activities *F* and *G* + Indirect cost for 8 months

$= 713 + (30 + 100 + 2 \times 30 + 2 \times 70) + 50 = \text{Rs } 1093$

Since the total project cost for 8 months is more than the cost for 10 months, therefore, further crashing is not desirable. Hence, optimum pair of time and cost associated with the project is 10 months and Rs 993, as shown in Table 13.15.

Table 13.15 Crashing Schedule of the Project

Project Duration (months)	*Crashing Activity with Months*	*Direct Cost (Rs '000)*			*Indirect Cost (Rs '000)*	*Total Cost (Rs '000)*
		Normal	*Crashing*	*Total*		
13	–	713	–	713	400	1,113
12	1 – 2(1)	713	30	743	250	993
10	2 – 4(2)	713	30 + 2 × 50 = 130	843	100	943
8	2 – 5(2) 4 – 5(2)	713	130 + 2 × 30 + 2 × 70 = 330	1,043	50	1,093

SELF PRACTICE PROBLEMS D

1. The following table gives the activities in a construction project and other relevant information:

Activity	*Description*	*Immediate Predecessor*	*Normal*		*Crash*		*Cost Slope*	*Rank*
			Time	*Cost*	*Time*	*Cost*		
A	Dismantle and inspect	–	2	100	1	140	40	*II*
B	Sub-contract repairs	*A*	6	200	2	500	75	*IV*
C	Buy – spares	*A*	3	100	1	140	20	*I*
D	Rebuilt/repair	*A*	4	500	2	800	150	*V*
E	Assemble	*B, C, D*	2	200	1	250	50	*III*
F	Trial run	*E*	1	50	–	250	–	–

Fixed-cost data for the different project durations are given below:

Duration (in days)	Cost in (Rs)
11	1,100
10	1,000
9	900
8	800
7	700
6	600
5	500
4	400

(a) Draw the network diagram for the project.
(b) Determine the project duration which will return in minimum total project cost.

2. The required data for a small project consisting of different activities are given below:

Activity	Predecessor Activities	Normal		Crash	
		Duration (days)	Cost (Rs)	Duration (days)	Cost (Rs)
A	–	6	300	5	400
B	–	8	400	6	600
C	A	7	400	5	600
D	B	12	1,000	4	1,400
E	C	8	800	8	800
F	B	7	400	6	500
G	D, E	5	1,000	3	1,400
H	F	8	500	5	700

(a) Draw the network diagram for the project and find the normal and minimum project length.
(b) If the project is to be completed in 21 days with minimum crash cost which activities should be crashed to how many days?

3. The time and cost estimates and precedence relationship of the different activities constituting a project are given below:

Activity	Predecessor Activities	Time (in weeks)		Cost (in Rs)	
		Normal	Crash	Normal	Crash
A	–	3	2	8,000	19,000
B	–	8	6	600	1,000
C	B	6	4	10,000	12,000
D	B	5	2	4,000	10,000
E	A	13	10	3,000	9,000
F	A	4	4	15,000	15,000
G	F	2	1	1,200	1,400
H	C, E, G	6	4	3,500	4,500
I	F	2	1	7,000	8,000

(a) Draw a project network diagram and find the critical path.
(b) If a dead line of 17 weeks is imposed for completion of the project, what activities will be crashed, what would be the additional cost and what would be critical activities of the crashed network after crashing?

4. The activities of a project are tabulated below with immediate predecessors and normal and crash time cost.

Activity	Immediate Predecessor	Normal		Crash	
		Cost (Rs)	Time (days)	Cost (Rs)	Time (days)
A	–	200	3	400	2
B	–	250	8	700	5
C	–	320	5	380	4
D	A	410	0	800	4
E	C	600	2	670	1
F	B, E	400	6	950	1
H	B, E	550	12	1,000	6
G	D	300	11	400	9

(a) Draw the project network diagram using normal time of activities.
(b) Determine the critical path and the normal duration and cost of the project.
(c) Suitably crash the activities so that the normal duration may be reduced by 3 days at minimum cost. Also find the project cost for this shortened duration if the indirect cost per day is Rs 25.

5. Patel Machinery Co. has been offered a contract to build and deliver nine extruding presses to the ABC Bottling Co. The contract price is contingent on meeting a specified delivery time, a bonus being given for early delivery. The marketing department has established the following cost and time information:

Activity	Normal Time (weeks)			Normal Cost (Rs)	Crash Time (weeks)	Crash Cost (Rs)
	Optimistic	Pessimistic	Most Likely			
1 – 2	1	5	3	15,000	1	19,000
2 – 3	1	7	4	18,000	3	24,000
2 – 4	1	5	3	14,000	2	16,000
2 – 5	5	11	8	15,000	7	16,000
3 – 6	2	6	4	13,000	2	15,000
4 – 6	5	7	6	12,000	4	13,000
5 – 7	4	6	5	20,000	4	24,000
6 – 7	1	5	3	17,000	1	20,000

The normal delivery time is 16 weeks for a contract price of Rs 1,24,000. Based on the probability for each of the following specified delivery time, recommend the delivery schedule that the Patel Machinery Co. should follow:

Contract Delivery Time (weeks)	Contract Amount (Rs)
15	1,42,500
14	1,45,000
13	1,50,000
12	1,52,500

6. The President of ABC Manufacturing Company has an opportunity to participate in a project that has a sales price of Rs 90,000 but the project must be completed within 8 weeks. This letter of intent was received on Friday afternoon. Both the superintendent of production and the cost accountant came in on Saturday and completed the appropriate time and cost for you based upon past jobs. Since the president needs an answer at 8.30 a.m. on Monday (start of the 8th week), you, have been requested to determine the profitability of the project on an 8 week basis. An answer at 8.30 a.m. Monday allows the firm to start the production order at 10.00 a.m. in order to stay within the 8 weeks demanded by the customer. The time and cost under normal conditions without crashing the project is based upon an 11 week basis. What answer should the president give the customer on Monday morning. A table of times and cost is given below:

Event	Preceding Event	Normal		Crash	
		Time (weeks)	Cost (Rs)	Time (weeks)	Cost (Rs)
4	1	2	8,000	1	13,000
2	1	2	7,000	1	19,000
3	1	6	11,000	5	13,000
4	2	4	6,000	3	10,000
3	2	2	9,000	1	10,000
5	2	7	8,500	6	11,000
5	4	4	10,000	3	16,000
5	3	3	5,000	2	7,000

7. The time-cost estimates for the various activities of a project are given below:

Activity	Preceding Activities	Time (weeks)		Cost (Rs)	
		Normal	Crash	Normal	Crash
A	–	8	6	8,000	10,000
B	–	7	5	6,000	8,400
C	A	5	4	7,000	8,500

(Conted..)

Activity	Preceding Activities	Time (weeks)		Cost (Rs)	
		Normal	Crash	Normal	Crash
D	B	4	3	3,000	3,800
E	A	3	2	2,000	2,600
F	D, E	5	3	5,000	6,600
G	C	4	3	6,000	7,000

The project manager wishes to complete the project in the minimum possible time. However, he is not authorized to spend more than Rs 5,000 on crashing.

Suggest the least-cost schedule for achieving the objective of the project manager. Assume that there is no indirect or utility cost. [*Delhi Univ., MCom, 2000*]

8. An electronics firm has signed a contract to install an instrument landing device at the local airport. The complete installation can be broken down into fourteen separate activities. Each activity (labelled A through N), its predecessor activities, normal time and cost and crash time and cost are given in the following table. The contract specifies that the installation will be completed within 18 days. There is a penalty of Rs 100 per day beyond the specified completion time.

Activity	Preceding	Normal Time (days)	Normal Cost (Rs)	Crash Time (days)	Crash Cost (Rs)
A	–	3	320	2	360
B	–	5	550	4	600
C	–	6	575	4	700
D	A	7	750	5	850
E	A	4	420	3	490
F	B, D	2	180	2	280
G	C	4	425	3	485
H	A	8	850	5	900
I	C	5	475	4	535
J	C	7	675	5	735
K	E, F, G	4	400	3	440
L	H, I	6	650	4	750
M	L	3	280	2	335
N	J, K	5	525	4	575

The project will require the positioning of one engineer till the completion of work. The monthly cost of each engineer for such job is Rs 10,000. This will be considered as indirect cost.

9. American Bottle Company (ABC) produces several types of glass containers. They have recently reduced capacity at several of their plants. The manufacturing of these grass containers involves large, expensive machines (including ovens), several of which were shut down during capacity reduction. The machines are hard to shut down and to start up. In the event of a surge in demand, they want to know how quickly they could start one back again. How quickly can they start a new oven using normal times? What is the fastest time in which a new oven can be started, and how much additional cost would this involve?

Cost (Rs) per Unit Time (hour) Reduction	Activity		Normal Time	Crash Time	Predecessor
–	A	Preheat glass	8	8	C
–	B	Preheat oven	12	12	D
400	C	Obtain materials	4	2	–
200	D	Check valves	4	2	–
200	E	Check pressure seals	2	1	B
–	F	Add glass to oven	2	2	A, E
500	G	Prepare bottlemaker	6	3	E
	H	Run test production	4	4	F, G
500	I	Examine test quantity and make adjustments	4	2	H
	J	Refill oven with glass	2	2	H

10. A company has recently won a contract for the installation of a die casting machine and its associated building construction work at a local factory of a large national firm of electronic engineers. The following table gives the various activities involved in this job, their normal time and cost estimates and their crash time and their cost estimates.

Activity	Description	Predecessor	Normal		Crash	
			Time (days)	Cost (Rs)	Time (days)	Cost (Rs)
A	Prepare foundations and underground services and erect building frame structure	–	30	90,000	25	1,05,000
B	Fabricate part and assemble steel frames to support the machine	–	25	1,80,000	20	1,90,000
C	Collect die casting machine and its associated gear from the manufacturers	–	10	50,000	8	54,000
D	Assemble and check control gear	C	10	7,500	7	9,000
E	Fit control gear on steel frames and instal	B, D	10	4,200	10	4,200
F	Fit aluminium sheet wall claddings	A, E	20	20,000	16	30,000
G	Erect assembled plant on to prepared foundation and framework and connect services	A, E	35	28,000	30	35,500
H	Erect mechanical handling plant	B, D	20	12,000	18	15,000
I	Fit ventilation and fire protection system	F	20	14,000	15	24,000

If the variable overhead costs are Rs 5,000 per day, determine the optimal project duration. [*Delhi Univ., MBA, Dec. 1995*]

11. A small marketing project consists of the jobs given in the table given below. Next to each job is listed its normal time and a minimum or crash time (in days). The cost (in Rs per day) of crashing each job is also given.

Job	Normal Duration (days)	Crash Duration (days)	Cost of Crashing (Rs per Day)
1 – 2	9	6	20
1 – 3	8	5	25
1 – 4	15	10	30
2 – 4	5	3	10
3 – 4	10	6	15
4 – 5	2	1	40

(a) What is the normal project length and the minimum project length?

(b) Determine the minimum crashing costs of schedules ranging from normal length, down to, and including, the minimum length schedule, i.e. if L is the length of the normal schedule, find the costs of schedules which are L, $L - 1$, $L - 2$ days long and so on.

The overhead cost for the project is Rs 60 per day. What is the optimal length schedule duration of each job for your solution?

12. The following table gives the list of various activities involved in the production of a wireless communication equipment, their immediate predecessor(s), their normal time and cost estimates and their crash time and cost estimates:

Activity	Description	Time (months)		Cost (Rs)		Predecessor
		Normal	Crash	Normal	Crash	
A	System calculations	3	1	45,000	63,000	–
B	Release of drawings	2	1	15,000	23,000	–
C	Procurement of PSU	12	10	85,000	1,05,400	A, B
D	Procurement of raw material	8	6	4,50,000	5,00,000	A, B
E	Production documentation	4	3	40,000	49,500	A, B
F	Time study/shop order	3	1	25,000	41,000	E
G	PCB Manufacture	6	5	50,000	59,000	D, F
H	Mechanical parts manufacturers	7	6	2,00,000	2,20,000	D, E, F
I	Electronic assembly	2	2	43,000	43,000	C, G, H
J	Testing	4	2	50,500	75,000	I

The indirect costs per month are Rs 20,000. Determine the optimal time versus the cost schedule.

13. The Sales Manager of Domestic Products Limited, Mumbai, was informed by the R&D department about the completion of the prototype of a particular product. He consulted the production manager on the time taken to produce the first batch of the product, which was needed for demonstration during his sales promotion programme. He also decided to invite a few industrial representatives to the demonstration of this new product and through them to launch it in the market. The various activities involved in this marketing project, their descriptions, estimated durations (in days) and immediate predecessors are given in the following tables:

Activity	Description	Duration (days)	Predecessor
A	Collect data on specifications and capabilities	4	–
B	Prepare operation manual	4	A
C	Chart out promotion programme	4	B
D	Make copies of manual and promotion material	9	B
E	Produce first batch for demonstration	16	B
F	Prepare list of press representatives	2	C
G	Chief executives conference with managers	1	C
H	Press representatives reach Bombay	2	F, G
I	Promotional meetings	4	D, H
J	Product demonstration	2	E, I
K	Press representatives return home	2	J

(a) Determine the maximum time required to complete the above project, list the critical activities and find the total float and free float, if any, for all the non-critical activities.

(b) If the indirect cost per day for the project under consideration is Rs 300, the normal and crash time and cost estimate for various activities are as given in the following table. Determine the optimal project duration.

Activity	Normal		Crash	
	Time (days)	Cost (Rs)	Time (days)	Cost (Rs)
A	4	100	3	450
B	4	150	2	510
C	4	200	4	200
D	9	500	4	1,000
E	16	2,000	8	2,960
F	2	60	1	140
G	1	100	1	100
H	2	2,500	1	6,000
I	4	2,200	3	2,340
J	2	700	2	700
K	2	2,500	1	6,000

HINTS AND ANSWERS

1.

Project Length	Critical Path(s)	Crashed Activities	Total cost: Direct + Increment + Indirect
10	A–B–F–G	A	1,150 + 1 × 40 + 1,000 = 2,190
9	A–B–F–G	F	1,190 + 2 × 50 + 900 = 2,140
7	A–B–F–G	B	1,140 + 2 × 75 + 700 = 2,090

2. (a) First crash activity *A* by one day at a cost of Rs 100 and then activity *G* at a cost of Rs 400.

(b) Crash activity *H* by 2 days at a cost of Rs 400/3, activity *C* by 2 days at a cost of Rs 200 and activity *D* by 2 days at a cost of Rs 400.

Total additional cost for minimum project duration of 21 days is Rs (100 + 400 + 400/3 + 200 + 400) = Rs 1,233.33.

3.

Project Length	Critical Path(s)	Crashed Activity	Total cost: Direct + Increment
21	A – E – H	H	52,300 + 1 × 500 = 52,800
20	A – E – H	H	52,800 + 1 × 500 = 53,300
19	A – E – H A – E – H	A	53,300 + 1 × 1,000 = 54,300
18	A – E – H B – C – H	E	54,300 + 1 × 2,000 = 56,300
17	A – E – H B – C – H	B, E	56,300 + 1 × 200 + 1 × 2,000 = 58,500

4. Critical path A – H – D. Reduce activity *H* by 2 days at a cost of Rs 100 and activity *D* by one day at a cost of Rs 78. Thus

Total cost = Direct cost + Increase in direct cost due to crashing + Indirect cost for 20 days project duration

= (3,030 + 2 × 50 + 1 times 78) + 20 × 25

= Rs 3,708.

5. Critical path: 1 – 2 – 5 – 7 with project duration of 16 weeks and cost of Rs 1,24,000.

Weeks	Contract Amount (Rs)	Contract Cost (Rs)	Project
16	1,24,000	1,24,000	–
15	1,42,000	1,25,000	17,500
14	1,45,000	1,27,000	18,000
13	1,50,000	1,29,000	21,000

13.8 UPDATING OF THE PROJECT PROGRESS

While executing a project, there are certain unexpected delays and difficulties in terms of supply of materials, availability of machines and/or breakdown of machines, availability of skilled manpower, natural calamity, etc. In such cases, it is necessary to review the progress of planning and scheduling to take stock of the progress that has been made. Such review is necessary to take remedial action in terms of time and resources required for the uncompleted activities in the project.

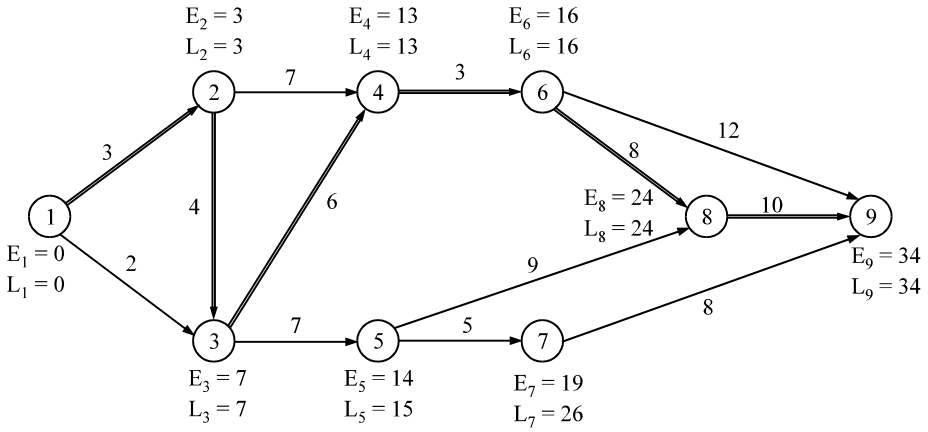
As such there is no rule about the specific time required to review the project progress. However, to add dynamism to the nature and progress of work, updating may be carried out as frequently as economically possible.

The project progress can be updated in two ways:

(i) Use the revised time estimate of incomplete activities and from the initial event calculate the earliest and latest completion time of each event in order to know the project completion time.

(ii) Change the complete work to zero duration and represent all the activities already finished by an arrow called the *elapsed time arrow.* Events in the revised network diagram are renumbered. The completion times of remaining activities are taken as the revised time.

Example 13.11 The network diagram for a project is shown below. A review of the project after 15 days reveals that:



(i) Activities 1 – 2, 1 – 3, 2 – 3, 2 – 4 and 3 – 4 are completed.

(ii) Activities 3 – 5 and 4 – 6 are in progress and need 2 and 4 days more, respectively.

(iii) The revised estimate shows that activity 8 – 9 will take only 8 days but 7 – 9 will need 10 days.

Draw a new network diagram after updating the project and determine the new critical path. The work completed may be shown by an elapsed time activity.

Solution The progress of the work noted at the end of 15th day from the start of the project may be summarized as shown in Table 13.16.

Table 13.16 Status of Progress

Activity	*Time Required (days)*	*Job Status*
1 – 2	0	Complete
1 – 3	0	Complete
2 – 3	0	Complete
2 – 4	0	Complete
3 – 4	0	Complete
3 – 5 (20 – 25)	2	In progress
4 – 6 (20 – 26)	4	In progress
5 – 8 (25 – 28)	9	Not started
5 – 7 (25 – 27)	5	Not started
6 – 8 (26 – 28)	8	Not started
6 – 9 (26 – 29)	12	Not started
7 – 9 (27 – 29)	10	Not started
8 – 9 (28 – 29)	8	Not started

In the network diagram, as shown in Fig. 13.28, activity 1 – 20 shows the elapsed time of 15 days. Other activities are assigned the time that is required for their completion after review. This is given in Table 13.16. Based on these revised time estimates, the critical path: 1 – 20 – 26 – 28 – 29 is shown by thick lines in Fig. 13.28. The total project duration has increased by one day.

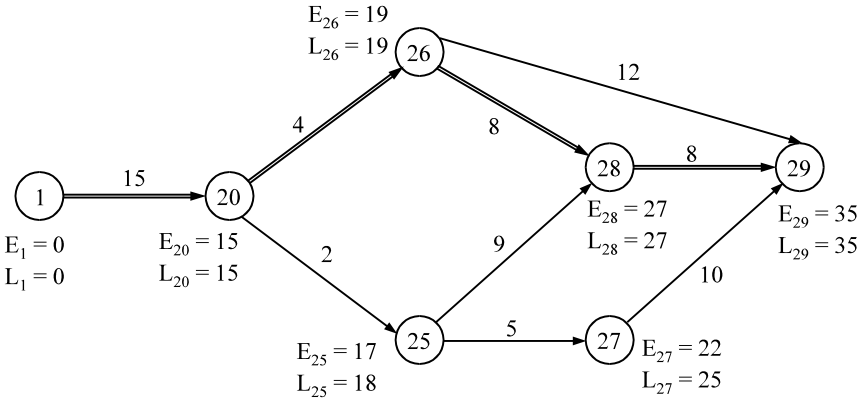


Fig. 13.28
Network Diagram

13.9 RESOURCE ALLOCATION

Resources such as men, money, material, machinery, etc., are limited and conflicting demands are made for the same type of resources as a project progresses. A systematic method for the allocation of resources therefore becomes essential. The aim is to prevent the day-to-day fluctuation in the level of the required resources and obtain a uniform resource requirement during the project duration. There are two approaches of optimum utilization of resources that are required to complete a project.

13.9.1 Resource Levelling

The analysis for stabilizing the rate of resource utilization by various activities at different times without changing the project duration, is called *resource levelling*.

The amount of total float of non-critical activities is used to stabilize the use of the existing level of resources. The resource requirement may be minimized by shifting a non-critical activity between its earliest start time and latest allowable time. The following two general rules are normally used in scheduling non-critical activities:

(i) If the total float of a non-critical activity is equal to its free float, then it can be scheduled anywhere between its earliest start and latest completion times.

(ii) If the total float of a non-critical activity is more than its free float, then its starting time can be delayed relative to its earliest start time by no more than the amount of its float, without affecting the scheduling of its immediately succeeding activities.

13.9.2 Resource Smoothing

The efforts to reduce the peak demand for resources and to reallocate them among the activities of a project so that the total project duration remains the shortest is known as *resource smoothing* (*or loading*).

The procedure of carrying out resource smoothing can be summarized in the following steps.

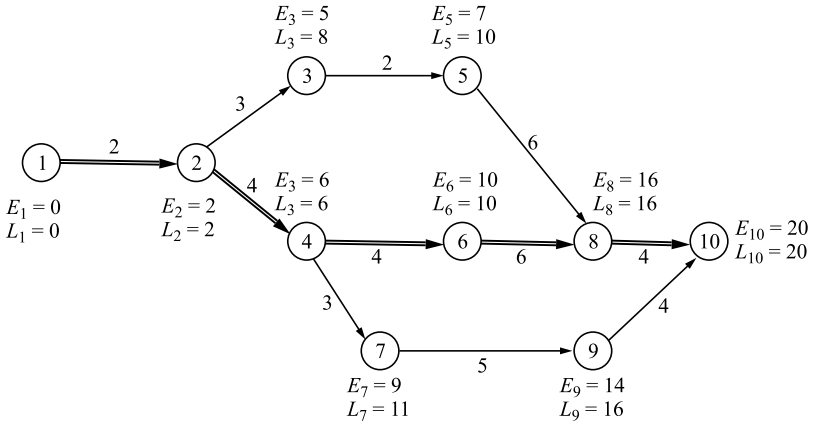
Step 1: Calculate the earliest start and latest finish times of each activity and then draw a time scaled version (or squared) of the network. A critical path is drawn along a straight line and non-critical activities on both sides of this line. Resource requirement of each activity is given along the arrows.

Step 2: Draw the resource histogram by taking earliest start times or latest start times of activities on the *x*-axis and required cumulative resource on *y*-axis.

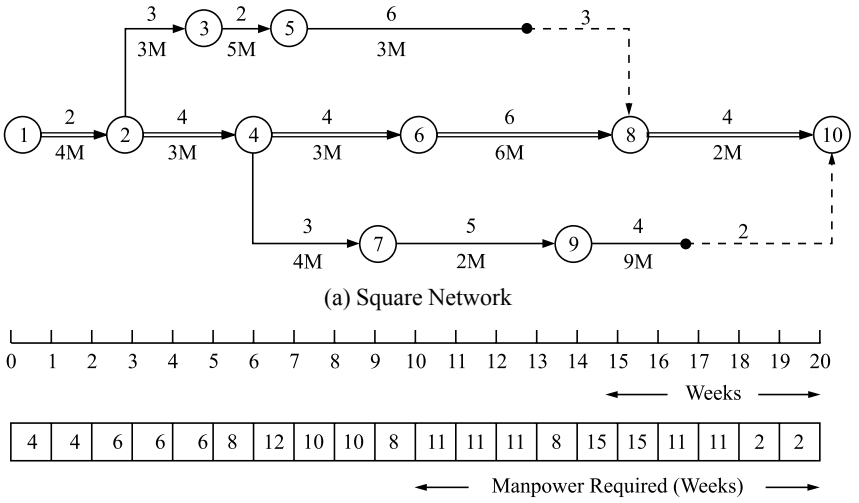
Step 3: Shift the start time of those non-critical activities that have the largest float in order to smoothen the resources.

Example 13.12 A network with the following activity durations and manpower requirement is given. Analyse the project form point of view of resource to bring out the necessary steps involved in the analysis and smoothing of resources.

Activity	:	1 – 2	2 – 3	2 – 4	3 – 5	4 – 6	4 – 7	5 – 8	6 – 8	7 – 9	8 – 10	9 – 10
Duration (weeks)	:	2	3	4	2	4	3	6	6	5	4	4
Manpower required	:	4	3	3	5	3	4	3	6	2	2	9



Solution The critical path is drawn through events where E-values and L-values are equal by a thick line. The critical path is: 1 – 2 – 4 – 6 – 8 – 10, with the project duration of 20 weeks. The network diagram in the time-scaled form, also called *squared network* is shown in Fig. 13.29 where critical path is drawn along a horizontal line. The dotted lines show the total float of an activity



(a) Square Network

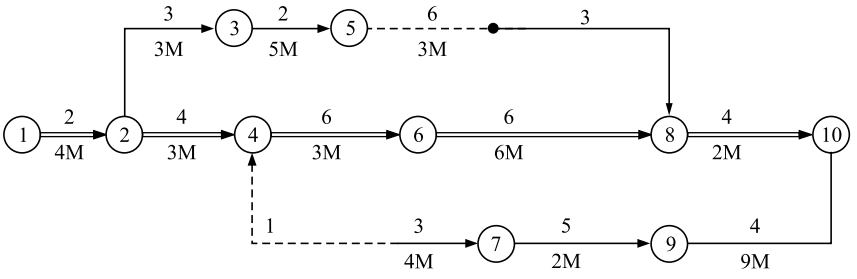
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

4	4	6	6	6	8	12	10	10	8	11	11	11	8	15	15	11	11	2	2

(b) Loard Chart

Fig. 13.29 Squared Network

The squared network is based on the earliest start times, and has been obtained by vertically summing up the manpower requirements for each week. The maximum demand of 15 men occurs in the 15th and 16th weeks.



(a) Square Network

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Weeks

4	4	6	6	6	8	8	3	7	7	9	11	11	11	11	11	11	11	11	11

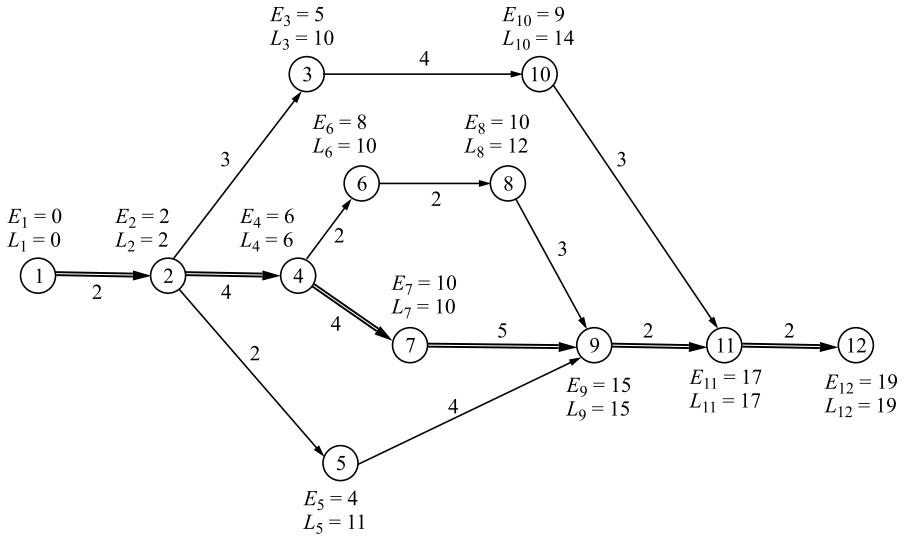
Manpower required weeks

(b) Loard Chart

Fig. 13.30 Resource Smoothing

As shown in Fig. 13.30, activities are shifted depending upon the floats. The activity path 4–7–9–10 has a float of two weeks, and the activities 7–8 and 9–10 are shifted to the right so that the start of each is delayed by two weeks. Similarly, activity 5–8 can be shifted to the right so that it starts on 11th day instead of starting on 8th day. The loading chart indicates that the maximum manpower required is 11 men. Thus, the project can be completed in 20 weeks by 11 men as compared to 15 men for the previous schedule.

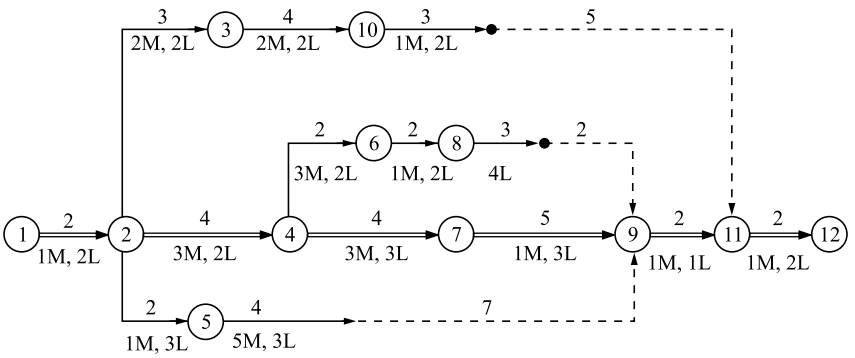
Example 13.13 Consider the following network diagram. Durations of the activities are marked along their arrows. The requirement of masons (M) and labourers (L) for each activity is given the table below. Analyse the project and smoothen the requirement of the resources.



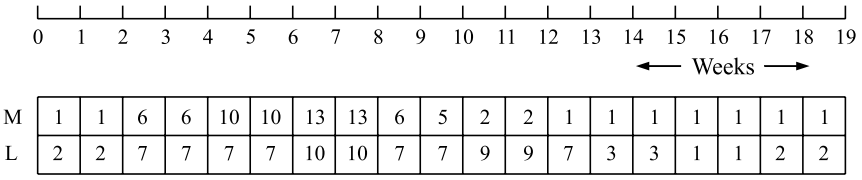
Activity (*weeks*)	*Duration* (*weeks*)	*Masons* (*M*)	*Laboures* (*L*)
1 – 2	2	1	2
2 – 3	3	2	2
2 – 4	4	3	2
2 – 5	2	1	3
3 – 10	4	2	2
4 – 6	2	3	2
4 – 7	4	3	3
5 – 9	4	5	3
6 – 8	2	1	2
7 – 9	5	1	3
8 – 9	3	–	4
9 – 11	2	1	1
10 – 11	3	1	2
11 – 12	2	1	2

Solution The critical path is identified by first calculating E-values and L-values and then joining those events by a thick line where these values are equal. The critical path is: 1 – 2 – 4 – 7 – 9 – 11 – 12 with the project duration of 19 weeks.

The network diagram is represented in the time scaled form as shown in Fig. 13.31(a). The critical path is drawn along a horizontal line. The dotted lines show the total float of an activity. Durations and requirements of masons (M) and labourers (L) for each activity are marked along the activity arrows. The loading chart shows the total numbers of masons and labourers required each day.



(a) Square Network



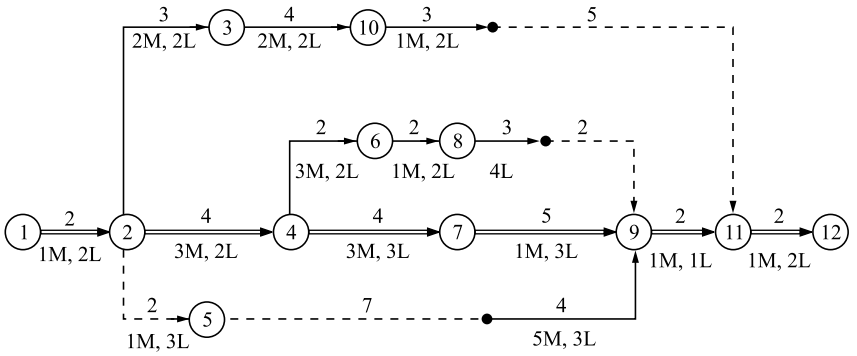
M	1	1	6	6	10	10	13	13	6	5	2	2	1	1	1	1	1	1	1
L	2	2	7	7	7	7	10	10	7	7	9	9	7	3	3	1	1	2	2

(b) Loard Chart

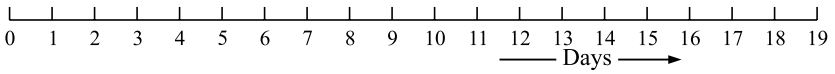
Fig. 13.31 Resource Smoothing

Figure 13.31(a) shows that activities 2–5 and 5–9 have a total float of 7 days. The start time of activity 5–9 can be shifted by 7 days so that it starts on 12th day instead of 5th day. Figure 13.32(a) represents the modified squared network along with the loading chart.

Figure 13.32(a) indicates that the demand for masons has decreased from 13 to 8 on the 7th and 8th days. However, the demand of labourers has increased from 9 to 12th day.



(a) Square Network



M	1	1	6	6	5	5	8	8	6	5	2	7	6	6	6	1	1	1	1
L	2	2	7	7	4	4	7	7	7	7	9	12	10	6	6	1	1	2	2

(b) Loard Chart

Fig. 13.32

As shown in Fig. 13.33(a), the start time of activity 8 – 9 can be shifted from 11th day to 13th day by utilising the float available for 2 days, wherein the requirement of labourers has also decreased from 12 to 10. The requirement of resources is smoothed without affecting the project duration.

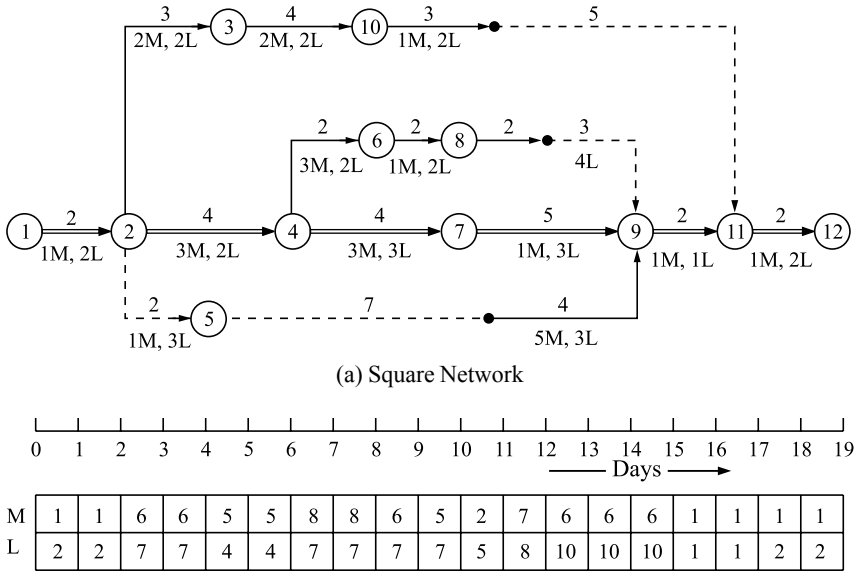


Fig. 13.33 Resource Smoothing

Example 13.14 Following are the manpower requirements for each activity in a project.

Activity	*Normal Time*	*Manpower Required*
0 – 1	2	4
1 – 2	3	3
1 – 3	4	3
2 – 4	2	5
3 – 5	4	3
3 – 6	3	4
4 – 7	6	3
5 – 7	6	6
6 – 8	5	2
7 – 9	4	2
8 – 9	4	9

(a) Draw the network diagram of the project activities.
(b) Rearrange the activities suitably for reducing the existing total manpower requirement.
(c) If only 9 men are available for the execution of the project, then rearrange the activities suitable for levelling the manpower resource.

Solution (a) The network diagram of the given project is shown in Fig. 13.34. Numbers in bracket with each activity in Fig. 13.34 is representing manpower requirement of the activity.

The earliest and latest completion times of each event has been calculated in the usual manner as described earlier and are indicated along the nodes in Fig. 13.34. The critical path: 0 – 1 – 3 – 5 – 7 – 9 is shown with thick lines. The total project duration is of 20 days.

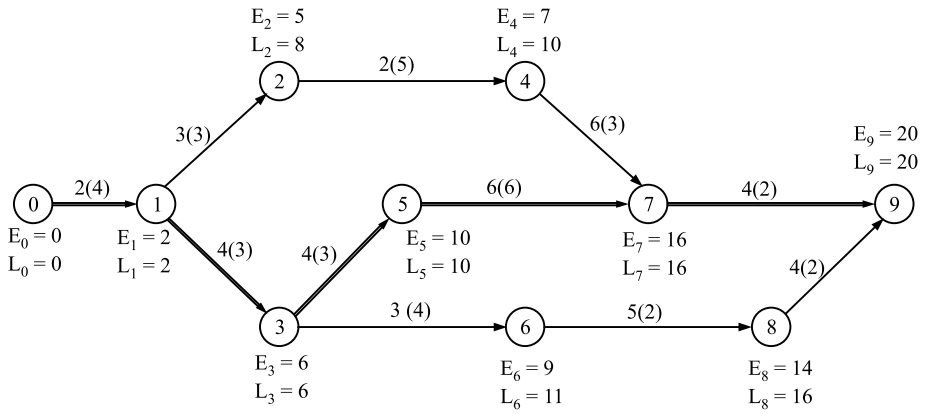


Fig. 13.34 Network Diagram

In order to exhibit activities on a time scale so that ranges of time allowable for various activities may be known, a time scaled version of the given network (also squared network) has been drawn. For this, first of all critical path is drawn along straight line and non-critical activities are added with it as shown in Fig. 13.35. Here, it is assumed that the various activities may be started as soon as possible, at their *earliest start* (E) times.

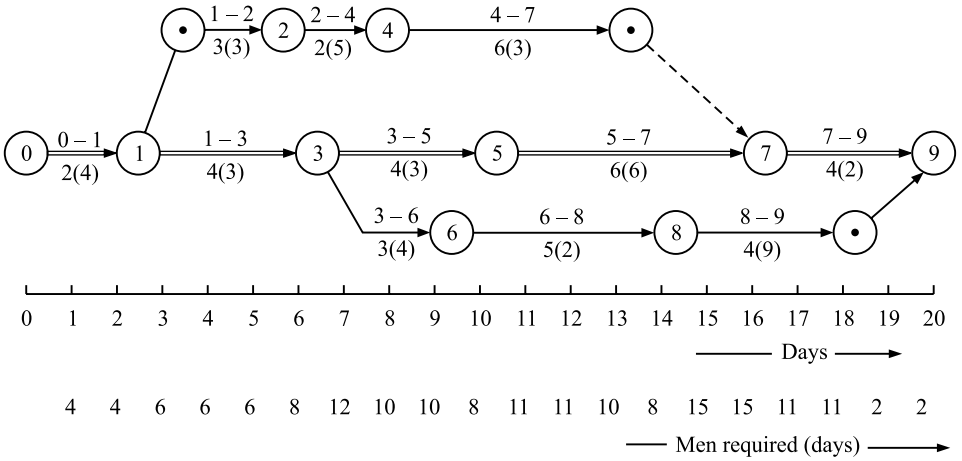


Fig. 13.35 Time Scaled Graph

The performance of some of the non-critical activities is shifted so that requirement of manpower may be rearranged. Because these non-critical activities have slack. To reduce the manpower requirements in the earlier weeks and to raise it in the last weeks, we may postpone the beginning of activity 3 – 6, 6 – 8 and 8 – 9 by two weeks each to start all at the beginning of 11th and 16th day, respectively and that of activity 4 – 7 by 3 days to start at the 10th day. The result of such rearrangement is shown in Fig. 13.36.

In Fig. 13.36, it may be observed that the requirement is less uneven in comparison with the previous situation and now 11 men are required compared to 15 men. Figures 13.35 and 13.36 indicate that there is no need of employing 11 men at a time. Men may be employed as per need.

Since only 9 men are available instead of 11, therefore activities have to be rearranged and expanded in terms of their completion time. This would increase the total project duration from 20 days to 25 days.

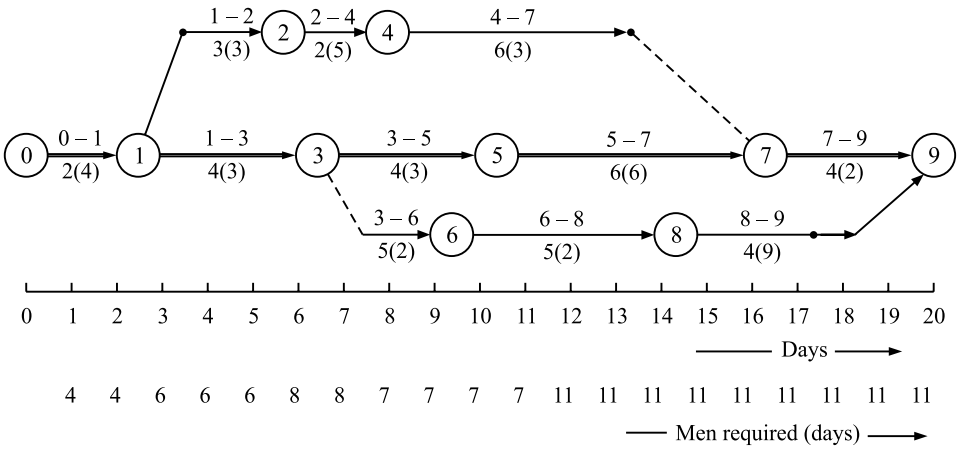


Fig. 13.36 Time Scaled Graph

Example 13.15 Following are the manpower requirement for each activity in a project.

Activity	*Normal Time (Days)*	*Manpower Required per Day*
1–2	10	2
1–3	11	3
2–4	13	4
2–6	14	3
3–4	10	1
4–5	7	3
4–6	17	3

5–7	13	5
6–7	9	8
7–8	1	11

(a) Draw the network and find out total float and free float for each activity.
(b) The contractor stipulates that during the first 26 days only 4 to 5 men and during remaining days 8 to 11 men only can be made available. Rearrange the activities suitably for levelling the manpower resources, satisfying the above condition.

Solution The network diagram of the given project is shown in Fig. 13.37. The value of total float and free float for non-critical activities are given in Table 13.16.

Activity	*Normal Time* (*Days*)	*Earliest Time*		*Latest Time*		*Total Float*	*Free Float*
		Start	*Finish*	*Start*	*Finish*		
1 – 2	11	0	11	2	13	2	0
2 – 6	14	10	24	26	40	16	16
3 – 4	10	11	21	13	23	2	2
4 – 5	7	23	30	29	36	6	0
5 – 7	13	30	43	36	49	6	6

Table 13.16 Activity Scheduling Times and Float

The total project duration is 50 days. Now it is required to rearrange activities so that they can be performed within 50 days (project duration) with the available manpower at different intervals.

The critical activities 1 – 2, 2 – 4, 4 – 6, 6 – 7 and 7 – 8 are scheduled first. Since non-critical activity 1 – 3 is scheduled at $t = 0$ followed by the non-critical activity 3 – 4, therefore, on 11th day (earliest finish time of 1 – 3) the manpower requirement would rise to 7, which is more than the available on 11th day. The total float of activity 1 – 3 is more than its free float, even than rule (b) is not applicable because its earliest start time as well as free float are both zero. Thus, it has to be scheduled otherwise it would result in a delay of the project.

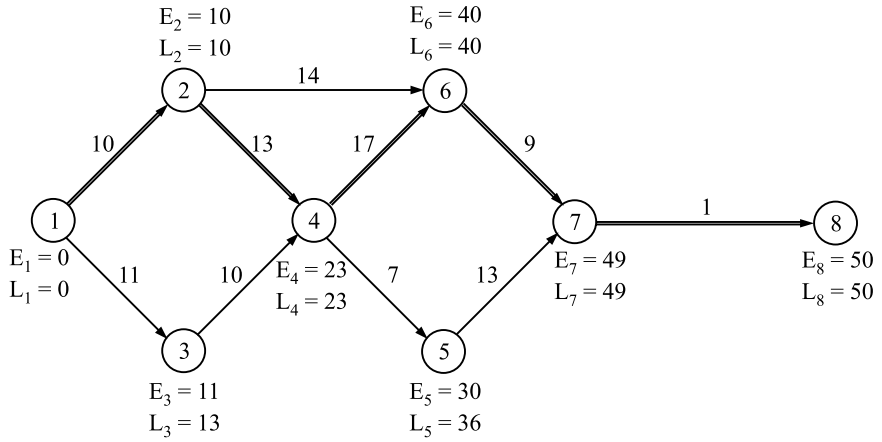


Fig. 13.37 Network Diagram

Starting time of non-critical activities 2 – 6 and 4 – 5 can be delayed both to $t = 26$ because more men would then be available (activity 4 – 6 can also be delayed upto $t = 29$). The last non-critical activity 5 – 7 is scheduled at $t = 33$. The total daily requirement of men is shown in Table 13.17 and manpower requirement of various activities is shown in bracket in Fig. 13.38.

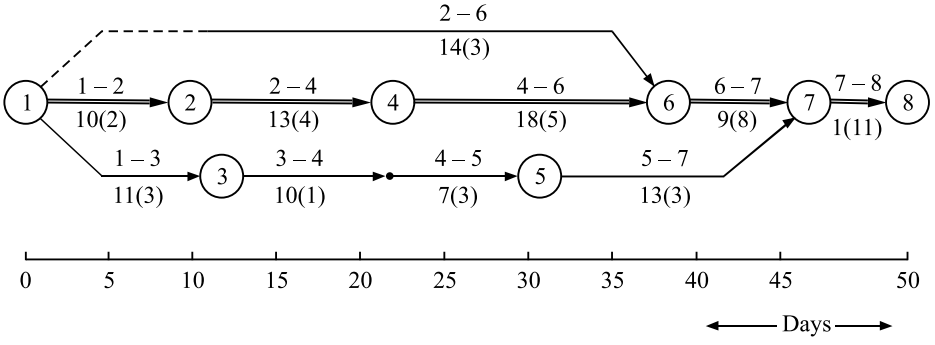


Fig. 13.38 Time Scaled Graph

Interval	*Day(s)*	*Men*
0 – 10	1 to 10th	5
10 – 11	11th	7
11 – 21	12th to 21st	5
21 – 23	22nd to 23rd	4
23 – 26	24th to 26th	5
26 – 46	27th to 46th	11
46 – 49	47th to 49th	8
49 – 50	50th	11

Table 13.17 Manpower Requirement

Example 13.16 A project with the following activities duration and manpower requirement is given:

Activity	:	1 – 2	1 – 3	1 – 4	2 – 5	2 – 6	3 – 7	4 – 8	5 – 9	6 – 9	7 – 8	8 – 9
Duration (days)	:	2	2	0	2	5	4	5	6	3	4	6
Manpower required	:	5	4	0	2	3	6	2	8	7	4	3

(a) Draw the network diagram of the project indicating the earliest start, earliest finish, latest finish and float of each activity.

(b) There are 11 persons who can be employed for this project. Carry out the appropriate manpower levelling so that the fluctuation of work force requirement from day-to-day is as small as possible.

Solution Network diagram based on activities schedule has been drawn as shown in Fig. 13.39. The critical path: 1 – 3 – 7 – 8 – 9 has been shown with thick lines.

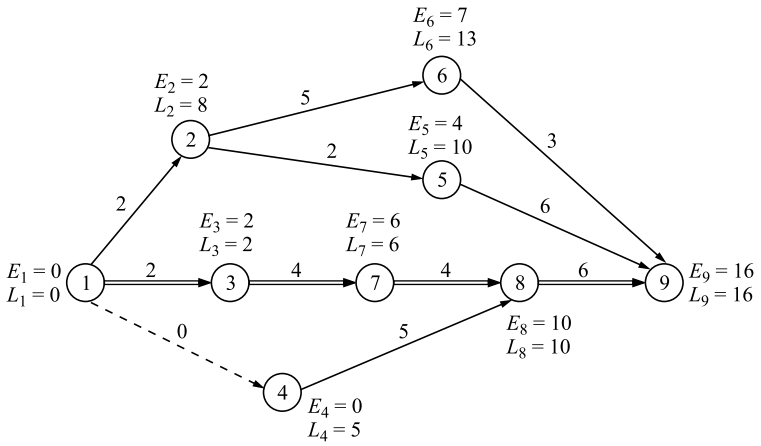


Fig. 13.39 Network Diagram

Table 13.18 illustrates man-days required for each activity and floats.

Activity	*Duration* (D)	*Manpower* (M)	*Man-days* ($M \times D$)	*Earliest Time*		*Latest Time*		*Total Float*
				Start (E_i)	*Finish* ($E_i + t_{ij}$)	*Start* ($L_j - t_{ij}$)	*Finish* (L_j)	$(L_j - t_{ij}) - E_i$
1 – 2	2	5	$5 \times 2 = 10$	0	2	6	8	6
1 – 3	2	4	$4 \times 2 = 8$	0	2	0	2	0
1 – 4	0	0	$0 \times 0 = 0$	0	0	5	5	5
2 – 5	2	2	$2 \times 2 = 4$	2	4	8	10	6
2 – 6	5	3	$3 \times 5 = 15$	2	7	8	13	6
3 – 7	4	6	$6 \times 4 = 24$	2	6	2	6	0
4 – 8	5	2	$5 \times 2 = 10$	0	5	5	10	5
5 – 9	6	8	$8 \times 6 = 48$	4	10	10	16	6
6 – 9	3	7	$7 \times 3 = 21$	7	10	13	16	6
7 – 8	4	4	$4 \times 4 = 16$	6	10	6	10	0
8 – 9	6	3	$3 \times 6 = 18$	10	16	10	16	0

Table 13.18

Table 13.19 represents the resource required to be allocated along with loaded chart as shown in Fig. 13.40. This figure indicates that five persons remain idle from 8th to 10th day and four persons remain idle from 17th to 19th day. The project cannot be completed in the normal time duration of 16 days and has to be delayed by 3 days if 11 persons only are to be employed.

Halting Time	*Available Resource*	*Activity*	*Activities in the Queue*			*Resource Allocated*
			Man days (M × D)	*Float*	*Priority*	
0	11 M	1 – 2	5 × 2 = 10	6	III	5 M
		1 – 3	4 × 2 = 8	0	I	4 M
		4 – 8	2 × 5 = 10	5	II	2 M
2	9 M	2 – 5	2 × 2 = 4	6	III	—
		2 – 6	3 × 5 = 15	6	II	3 M
		3 – 7	6 × 4 = 24	0	I	6 M
5	2 M	2 – 5	2 × 2 = 4	3	I	2 M
6	6 M	7 – 8	4 × 4 = 16	0	I	4 M
7	5 M	5 – 9	8 × 6 = 48	3	I	—
		6 – 9	7 × 3 = 21	6	II	—
		5 – 9	8 × 6 = 48	0	I	8 M
10	11 M	6 – 9	7 × 3 = 21	3	III	—
		8 – 9	3 × 6 = 18	0	II	3 M
16	11 M	6 – 9	7 × 3 = 21	–	I	7 M

Table 13.19 Resource Allocation

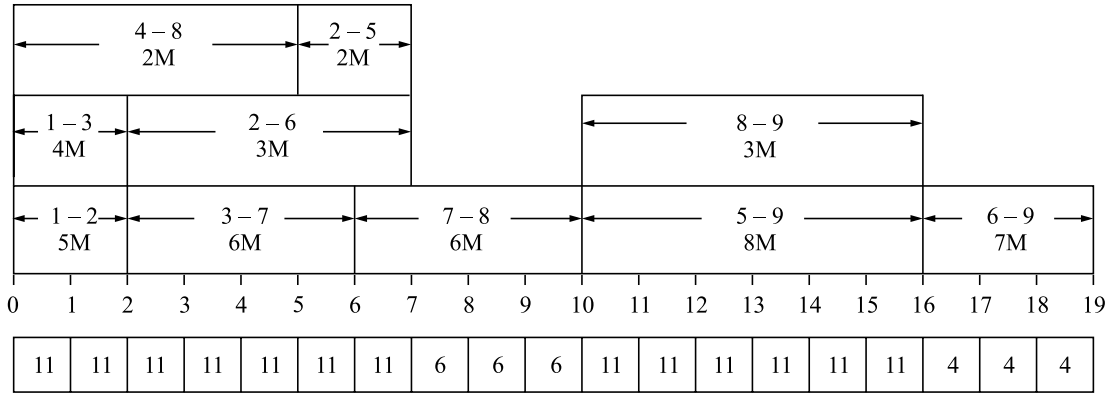


Fig. 13.40 Loading Chart

Example 13.17 For a project consisting of several activities, the durations and required resources for carrying out each of the activities and their availabilities are given below:

Activity	*Equipment*	*Operators*	*Duration (days)*
1 – 2	*X*	30	4
1 – 3	*Y*	20	3
1 – 4	*Z*	20	6
2 – 4	*X*	30	4
2 – 5	*Z*	20	8
3 – 4	*Y*	20	4
4 – 5	*Y*	20	4
4 – 5	*X*	30	6

Resources availability: (a) Operators = 50 (b) Equipment X = 1, Y = 1 and Z = 1.

(a) Draw the network, identify critical path and compute the total float for each of the activities.

(b) Find the project completion time under the given resource constraints.

Solution Network diagram based on activities schedule has been shown in Fig. 13.41. The critical path: 1 – 2 – 4 – 5 has been shown with thick lines

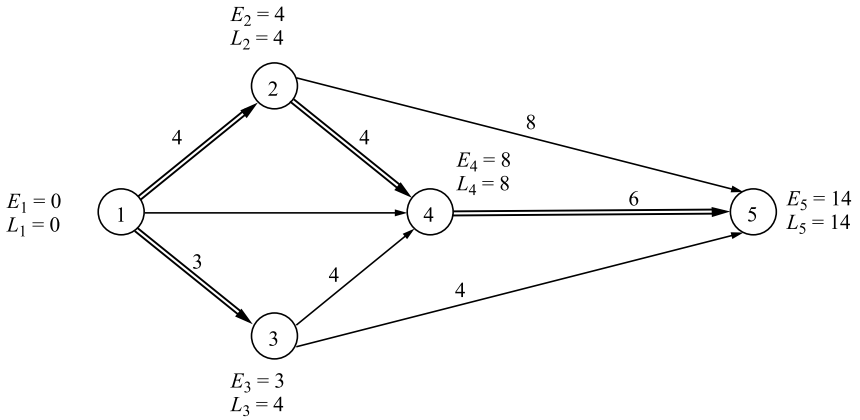


Fig. 13.41
Network Diagram

Table 13.20 illustrates man-days required for each activity and floats occurred.

Activity	*Duration* (D)	*Manpower* (M)	*Man-days* ($M \times D$)	*Equip-ment*	*Earliest Time*		*Latest Time*		*Total Float*
					Start (E_i)	*Finish* ($E_i + t_{ij}$)	*Start* ($L_j - t_{ij}$)	*Finish* (L_j)	$(L_j - t_{ij}) - E_i$
1 – 2	4	30	30 × 4 = 120	X	0	4	0	4	0
1 – 3	3	20	20 × 3 = 60	Y	0	3	1	4	0
1 – 4	6	20	20 × 6 = 120	Z	0	6	2	8	0
2 – 4	4	30	30 × 4 = 120	X	4	8	4	8	0
2 – 5	8	20	20 × 8 = 160	Z	4	12	6	14	0
3 – 4	4	20	20 × 4 = 80	Y	3	7	4	8	0
3 – 5	4	20	20 × 4 = 80	Y	3	7	10	14	6
4 – 5	6	30	30 × 6 = 180	X	8	10	8	14	2

Table 13.20
Network Analysis

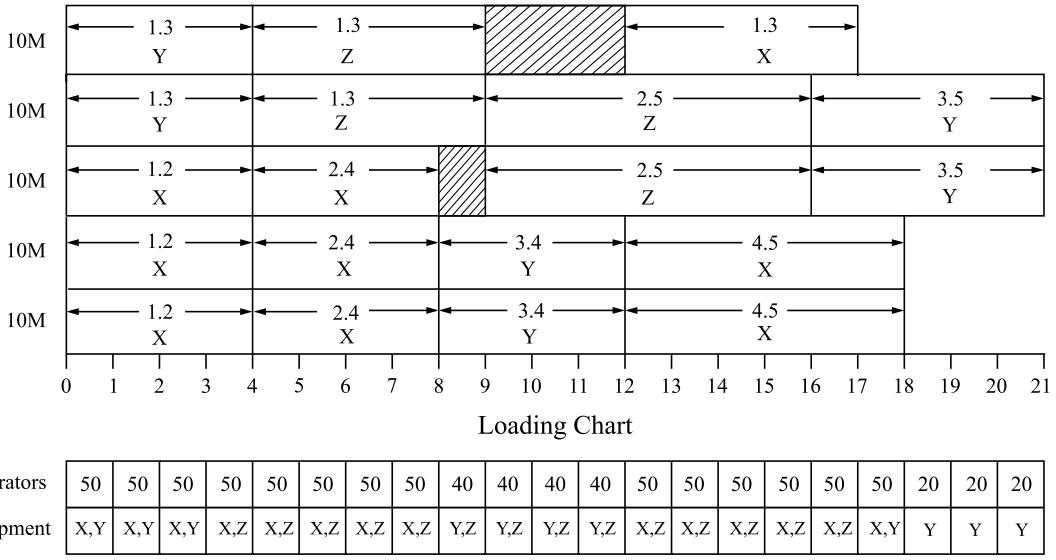
Table 13.20 represents the resource required to be allocated along with loaded chart as shown in Fig. 13.41. Activities 1–2, 1–3 are taken up at time 0 while 1 – 4 is delayed to start at time 3. Float of activity 1 – 4 becomes negative and therefore float will not be the criterion for finding priorities. At time 3, activities 3 – 4 and 3–5 have the same man-days, manpower size and activity sequence, therefore, priority II is assigned to activity 3 – 4 and priority III to activity 3–5.

At time 4, available resources are 30M, X and Y. Activity 2–5 has the highest man-days and is assigned priority I. However, since it requires equipment Z, it cannot be taken up and, instead, activity 2–4 with priority II is selected for execution. Figure 13.42 indicates that the project requires 21 days for completion and is, therefore, delayed by 7 days beyond normal completion time. The idle man-days are shown shaded. It may be observed that during 19th, 20th and 21st days only 20 persons are required. The remaining 30 persons can be relieved/shifted to other project after 18 days. Daily requirement of operators and equipment is shown in Fig. 13.42.

Halting Time	*Available Resources*	*Activity*	*Equipment*	*Man-days* ($M \times D$)	*Float*	*Priority*	*Resources Allocated*
0	50 M; X, Y, Z	1–2	X	30 × 4 = 120	0	I	30 M, X
		1–3	Y	20 × 3 = 60	I	II	20 M, Y
		1–4	Z	20 × 6 = 120	2	III	
3	20 M; Y, Z	1–4	Z	20 × 6 = 120	–	I	20 M, Z
		3–4	Y	20 × 4 = 80	–	II	
		3–5	Y	20 × 4 = 80	–	III	
4	30 M; X, Y	2–4	X	30 × 4 = 120	–	II	30 M, X
		2–5	Z	20 × 8 = 160	–	I	
		3–4	Y	20 × 4 = 80	–	III	
		3–5	Y	20 × 4 = 80	–	IV	
8	30 M; X, Y	2–5	Z	20 × 8 = 160	–	I	–
		3–4	Y	20 × 4 = 80	–	II	20 M, Y
		3–5	Y	20 × 4 = 80	–	III	

9	30 M; X, Z	2–5	Z	20 × 8 = 160	–	I	20 M,Z
		3–5	Y	20 × 4 = 80	–	II	
12	30 M; X, Y	3–5	Z	20 × 4 = 80	–	II	
		4–5	X	30 × 6 = 180	–	I	30 M, X
17	20 M; Y, Z	3–5	Y	20 × 4 = 80	–	I	20 M, Y

Table 13.21 Resource Allocation



Operators	50	50	50	50	50	50	50	50	40	40	40	40	50	50	50	50	50	50	20	20	20
Equipment	X,Y	X,Y	X,Y	X,Z	X,Z	X,Z	X,Z	X,Z	Y,Z	Y,Z	Y,Z	Y,Z	X,Z	X,Z	X,Z	X,Z	X,Z	X,Y	Y	Y	Y

Fig. 13.42 Loading Chart

SELF PRACTICE PROBLEMS E

1. The following table gives, for each activity of a project, its duration and corresponding resource requirements as well as the availability of each type of resource.

Activity	*Duration* (*days*)	*Resource Required*	
		Machines	*Men*
1 – 2	7	2	20
1 – 3	7	2	20
2 – 3	8	3	30
2 – 4	6	4	30
3 – 6	9	2	20
4 – 5	3	2	20
5 – 6	5	2	20
Minimum available resources		4	40

   (a) Draw the network, compute the earliest occurrence time, and the latest occurrence time for each event, the total float for each activity and identify the critical path, assuming that there are no resource constraints.
   (b) Under the given resource constraints find out the minimum duration to complete the project and compare the utilization of the resources for that duration.

2. The activities involved in a certain project have been identified as follows:

Activity	*Preceding Activity*	*Duration* (*weeks*)	*No. of Men Required*
A	–	4	1
B	–	7	1
C	–	8	2
D	*A*	5	3
E	*C*	4	1
F	*B, E*	4	2
G	*C*	11	2
H	*G, Fs*	4	1

   (a) For the above project draw the network. Determine the critical path and its duration.
   (b) If there were only three men available at any one time how long would the project take and how would you allocate the men to the activities?
   (c) If there were no restrictions on the amount of labour available, explain how you would schedule the activities.

3. The activities, activity durations and manpower requirements of a project are given below.

Activity	*Duration* (*days*)	*No. of Men Required*
1–2	2	5
1–3	2	4
1–4	0	0
2–5	2	2
2–6	5	3
3–7	4	6
4–8	5	2
5–9	6	8
6–9	3	7
7–8	4	4
8–9	6	3

There are eleven persons who can be employed for this project. Carry out the appropriate manpower levelling so that the fluctuations of workforce requirement from day-to-day are as small as possible.

4. A project has the following activities and their durations.

Activity	*Duration (days)*
12	13
13	15
14	9
24	10
25	27
34	7
36	18
47	30
57	12
67	10
68	10
78	9

(a) Draw the network of the project and find its duration.
(b) At the end of 25 days it is observed that
   (i) activities 12, 13, 14 have been completed.
   (ii) activity 24 is in process and will be completed in 5 more days.
   (iii) activity 36 is in progress and will need 20 more days for completion.
   (iv) activity 67 is presenting some problem and will take 15 days.

Draw the updated network and find out its revised duration. Number 12 denotes activity 1 – 2 and same is true for other activities.

5. A project has the following activities and their durations:

Activity	*Time (days)*	*Preceding Activity*
A	1	–
B	2	–
C	2	–
D	2	*A, B*
E	4	*B, C*
F	1	*C*
G	4	*D*
H	8	*G, E, F*

(a) Draw the project network and indicate the critical path.
(b) What is the minimum completion time?
(c) During the second day of work it is discovered that activity *F* would take 4 days instead of 1. Will this delay the project? If this activity takes 6 days, will the project be delayed?
(d) The company has limited number of men available to work on the project. Only two activities can be under way at the same time. Will this delay the project when compared to the time that unlimited resources would have been? (Activity *F* takes 6 days to complete).

6. The following table gives the activities in a small project and other relevant information:

Activity	*Duration (days)*	*Immediate Predecessor*	*Resource Required*	
			Operators	*Mechanics*
A	3	–	2	–
B	2	*A*	2	2
C	4	*A*	4	4
D	6	*A*	5	5
E	3	*B*	2	2
F	2	*E*	2	–
G	6	*C*	–	2
H	4	*D*	2	2
I	4	*G*	4	2
J	2	*D*	2	–
K	2	*J*	2	2
L	4	*F, H, I*	4	4

(a) Draw the network, compute the earliest start time and latest finish time for each of the activities and find out the project completion time. Also identify the critical path.
(b) Draw the time-scaled diagram with resource accumulation table. Comment on the demand for the operations and mechanics for the entire project duration and suggest the method of smoothening the resources.

CHAPTER SUMMARY

Ever since their inception in the late 1950s, PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method) have been used extensively to assist project managers in planning, scheduling, and controlling their projects.

The application of PERT/CPM begins by breaking down the project into its individual activities, identifying the immediate predecessors of each activity, and estimating the duration of each activity. The next step is to construct a project network to display this information.

PERT/CPM generate scheduling information for the project, including the earliest start time, the latest start time, and the slack for each activity. It also identifies the critical path of activities such that any delay along this path will delay project completion. Since the critical path is the longest path through the project network, its length determines the duration of the project, assuming all activities remain on schedule.

However, it is difficult for all activities to remain on schedule because there is often a considerable uncertainty about what the duration of an activity will turn out to be. The three-estimate approach in PERT deals with this situation by obtaining three different types of estimates (most likely, optimistic, and pessimistic) for the duration of each activity. This information is used to approximate the mean and variance of the probability distribution of this duration. It is then possible to approximate the probability that the project will be completed by the deadline.

The time-cost trade-offs approach in CPM enables the project manager to investigate the effect on total cost of changing the estimated duration of the project to various alternative values. The data needed for this activity are the time and cost for each activity when it is done in the normal way and then when it is fully crashed (expedited).

CHAPTER CONCEPTS QUIZ

True or False

1. The objective of network analysis is to minimize total project cost.
2. A project is an endeavour to create a unique product or service.
3. The techniques of operations research used for planning, scheduling and controlling projects are referred to as network analysis.
4. The CPM is sued for completing the projects that involves activities of repetitive nature.
5. PERT is referred to as an activity oriented technique.
6. PERT is a tool for planning and control of time.
7. Project planning phase allocates resources to work packages.
8. Scheduling phase identify manpower that will be responsible for each task.
9. A network in which the activities are represented by an arrow is referred to as activity-on-node network.
10. Beta probability distribution is often used in computing the expected activity completion times and variances in networks.

Fill in the Blanks

11. Resource leveling is the process of _____ the utilization of resources in a project.
12. Crashing is the process of reducing the total time that it takes to complete a project by expending _____.
13. _____ is the time consuming job or task that is a key subpart of the total project.
14. Earliest finish time that an activity can be finished without _____ of precedence requirements.
15. _____ is the point in time that marks the beginning or ending of an activity.
16. A bar chart indicating when the activities in a project will be performed is referred to as _____.
17. Network is the graphical display of a project that contains both _____ and _____.
18. A small circle or rectangle that is known as _____ serves as a junction point in the project network.
19. Latest finish time that an activity can be finished without _____ the entire project.
20. The amount of time that is expected to complete the activity is called _____.

Multiple Choice

21. The objective of network analysis is to
    (a) minimize total project duration
    (b) minimize total project cost
    (c) minimize production delays, interruption and conflicts
    (d) all of the above
22. Network models have advantage in terms of project
    (a) planning (b) scheduling
    (c) controlling (d) all of the above
23. The slack for an activity is equal to
    (a) LF – LS (b) EF – ES
    (c) LS – ES (d) none of the above
24. The another term commonly used for activity slack time is
    (a) total float (b) free float
    (c) independent float (d) all of the above
25. Generally the PERT technique deals with the project of
    (a) repetitive nature (b) non-repetitive nature
    (c) deterministic nature (d) none of the above
26. In PERT the span of time between the optimistic and pessimistic time estimates of an activity is
    (a) 3σ (b) 6σ
    (c) 12σ (d) none of the above
27. If an activity has zero slack, it implies that
    (a) it lies on the critical path (b) it is a dummy activity
    (c) the project is progressing well (d) none of the above
28. A dummy activity is used in the network diagram when
    (a) two parallel activities have the same tail and head events
    (b) the chain of activities may have a common event yet be independent by themselves
    (c) both (a) and (b)
    (d) none of the above
29. While drawing the network diagram, for each activity project, we should look
    (a) what activities precede this activity?
    (b) what activities follow this activity?
    (c) what activities can concurrently take place with this activity?
    (d) all of the above
30. In the PERT network each activity time assumes a Beta-distribution because
    (a) it is a unimodal distribution that provides information regarding the uncertainty of time estimates of activities
    (b) it has got finite non-negative error
    (c) it need not be symmetrical about model value
    (d) all of the above
31. The critical path satisfy the condition that
    (a) $E_i = L_i$ and $E_j = L_j$
    (b) $L_j - E_i = L_i - L_j$
    (c) $L_j - E_i = L_i - E_j = d$ (constant)
    (d) all of the above
32. Float or slack analysis is useful for
    (a) projects behind the schedule only
    (b) projects ahead of the schedule only
    (c) both a and b
    (d) none of the above
33. The activity that can be delayed without affecting the execution of the immediate succeeding activity is determined by
    (a) total float (b) free float
    (c) independent float (d) none of the above
34. In time cost-trade-off function analysis
    (a) cost decreases linearly as time increases
    (b) cost at normal time is zero
    (c) cost increases linearly as time increases
    (d) none of the above.
35. Activity-on-Arrow (AOA) diagram is preferred over Activity-on-Node (AON) diagram because
    (a) AOA diagrams are simple to construct
    (b) AOA diagrams give a better sense of the flow of time throughout a project
    (c) AOA diagrams do not involve dummy activities.
    (d) all of the above

Answers to Quiz

1. F	2. T	3. T	4. T	5. F	6. T	7. T	8. T	9. F	10. T
11. smoothing out		12. additional funds		13. activity		14. violation		15. event	
16. gantt chart		17. activities, events		18. node		19. delaying		20. most likely time	
21. (a)	22. (d)	23. (c)	24. (d)	25. (d)	26. (b)	27. (a)	28. (c)	29. (d)	30. (a)
31. (a)	32. (a)	33. (b)	34. (a)	35. (b)					

CASE STUDY

Case 13.1: Kaushik Mills

Kaushik Mills decided to increase its production capacity by building a new feed mill. The project consisted of a number of separate tasks, some of which could not be started before others were complete. Exhibit 1 lists the activities, along with the times expected for each, as 'agreed upon by management and the precedence relationships'.

The management wanted to advance the schedule as much as possible in order to save valuable time in getting the new mill into operation. The president of the mills commented: "Every week saved is worth Rs 70,000 in lost contribution if we can get going."

Some of the construction activities could be sped up. For example, the firm's architects could by working overtime, design the new plant in 10 weeks instead of the originally estimated 12 weeks. This advancement would cost the mills an additional Rs 25,000 per week that is advanced. The following table shows the maximum amount each activity could be crashed as well as the crash cost per week.

The president of mills had already held discussions with mill contractors, an independent firm which was a potential contractor for one of the projects major tasks, for building the plant. Kaushik Mills intended to do the other tasks either itself or through its agents. During the talks with mill contractors, the management had explored a number of bonus and penalty clauses. One of these was that for every week that the plant was built ahead of 10 weeks, the mills would pay contractors an additional Rs 75,000.

Activity	*Description Time (weeks)*	*Expected Activities*	*Precedent Time (weeks)*	*Minimum (Rs/week)*	*Crash*
A	Degin plant	12	–	10	24,000
B	Select plant site	8	*A*	18	–11
C	Select plant builder	6	*A*	11	3,000
D	Select operating personnel	13	*A*	13	–11
E	Prepare the building site	4	*B, C*	4	–11
F	Make or buy mill equipment	101	*C*	8	30,000
G	Prepare mill operations manual	6	*C*	4	1500
H	Build the plant	101	*E, F*	6	75,000
I	Train plant operators	8	*D, G*	8	–11
J	Test the plant	8	*H, I*	8	–11
K	Obtain a production licence	4	*J*	4	–11

The management of Kaushik Mills desires to know which activities it would crash and how it should schedule its workers.

Case 13.2: Krishna Foods

Krishna Foods has broken down the process of launching a new Fruit Yogot in the market into several steps. Some of these steps cannot begin until the others are completed. These relationships are shown in the following table. The estimates of the completion time (in weeks) of each activity and its immediate predecessor(s) are listed in the following table:

Activity	*Description*	*Predecessor*	*Time (weeks)*		
			Optimistic	*Likely*	*Pessimistic*
A	Management approval	–	2	2.5	4
B	Product concept test	*A*	3	4.7	5
C	Technical feasibility	*A*	2	2	3
D	Recipe finalization	*C, B*	1	1	2
E	Shelf life trials	*D*	8	12	15
F	Brand positioning study	*B*	4	5	7
G	Packaging key lines	*F*	1	1	2.5
H	Agency advertisement development	*F, E*	4	8	10
I	Agency: layouts artworks	*G*	2	3	5
J	Advertisement test research	*H*	3	4	5
K	Cost finalization	*D*	1	1	2
L	Pricing decision	*J, F*	1	1.5	3
M	Marketing mix finalization	*L*	2	2.5	3
N	POS development	*M*	2	4	5
O	Launch plans	*M*	1	1	1.5
P	Branch communication	*O*	0.5	0.5	1
Q	Supplier's delivery of packaging	*H*	4	5	8
R	Production trail	*D, Q*	1.5	2	3
S	Management final approval	*R*	0.5	1	1.5
T	Final production	*S*	1	2.5	3
U	Stock movement	*T*	0.5	1.5	2
V	Position movement	*T*	0.5	1	1.5
W	Launch	*U, V*	1	2	4

The management of the company desires to budget an amount for labour costs that will be sufficient with 95 per cent probability of completing the project. How much delay is possible in the project completion.